БИОЛОГИЯ · 10 КЛАСС · УРОК 13

Биотехнология

Основа: Biology Grade 10, Астана-кітап, 2019. Далее в этом PDF приложены все оригинальные страницы главы, включая рисунки, задания и таблицы. Русский раздел — учебный навигатор и проверочные задания, а не дословный перевод оригинала.

Как заниматься

Изучайте каждый подраздел на оригинальных страницах, используйте словарь терминов в учебнике, затем письменно отвечайте на пять вопросов. После всех подразделов выполните итоговую проверку из 40 вопросов. Проходной порог: 28 правильных ответов.

13.1 Микроорганизмы в повседневной жизни

Откройте раздел 13.1 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Микроорганизмы в повседневной жизни» и почему это важно.

1. Дайте определение понятия «Микроорганизмы в повседневной жизни».
2. Какую роль играет «Микроорганизмы в повседневной жизни» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Микроорганизмы в повседневной жизни»?
4. Приведите биологический пример, относящийся к теме «Микроорганизмы в повседневной жизни».
5. Как можно объяснить или проверить одно из утверждений из раздела «Микроорганизмы в повседневной жизни»?

13.2 Микроорганизмы в промышленности и медицине

Откройте раздел 13.2 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Микроорганизмы в промышленности и медицине» и почему это важно.

1. Дайте определение понятия «Микроорганизмы в промышленности и медицине».
2. Какую роль играет «Микроорганизмы в промышленности и медицине» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Микроорганизмы в промышленности и медицине»?
4. Приведите биологический пример, относящийся к теме «Микроорганизмы в промышленности и медицине».
5. Как можно объяснить или проверить одно из утверждений из раздела «Микроорганизмы в промышленности и медицине»?

13.3 Полимеразная цепная реакция

Откройте раздел 13.3 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Полимеразная цепная реакция» и почему это важно.

1. Дайте определение понятия «Полимеразная цепная реакция».
2. Какую роль играет «Полимеразная цепная реакция» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Полимеразная цепная реакция»?
4. Приведите биологический пример, относящийся к теме «Полимеразная цепная реакция».
5. Как можно объяснить или проверить одно из утверждений из раздела «Полимеразная цепная реакция»?

13.4 Применение ПЦР

Откройте раздел 13.4 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Применение ПЦР» и почему это важно.

1. Дайте определение понятия «Применение ПЦР».
2. Какую роль играет «Применение ПЦР» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Применение ПЦР»?
4. Приведите биологический пример, относящийся к теме «Применение ПЦР».
5. Как можно объяснить или проверить одно из утверждений из раздела «Применение ПЦР»?

13.5 Генная инженерия

Откройте раздел 13.5 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Генная инженерия» и почему это важно.

1. Дайте определение понятия «Генная инженерия».
2. Какую роль играет «Генная инженерия» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Генная инженерия»?
4. Приведите биологический пример, относящийся к теме «Генная инженерия».
5. Как можно объяснить или проверить одно из утверждений из раздела «Генная инженерия»?

13.6 Применение генной инженерии

Откройте раздел 13.6 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Применение генной инженерии» и почему это важно.

1. Дайте определение понятия «Применение генной инженерии».
2. Какую роль играет «Применение генной инженерии» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Применение генной инженерии»?
4. Приведите биологический пример, относящийся к теме «Применение генной инженерии».
5. Как можно объяснить или проверить одно из утверждений из раздела «Применение генной инженерии»?

13.7 Этика применения ГМО

Откройте раздел 13.7 в приложенных оригинальных страницах. Найдите определения, главные схемы и примеры. Сформулируйте своими словами, что означает «Этика применения ГМО» и почему это важно.

1. Дайте определение понятия «Этика применения ГМО».
2. Какую роль играет «Этика применения ГМО» в живых организмах?
3. Какие ключевые структуры, этапы или компоненты связаны с темой «Этика применения ГМО»?
4. Приведите биологический пример, относящийся к теме «Этика применения ГМО».
5. Как можно объяснить или проверить одно из утверждений из раздела «Этика применения ГМО»?

ИТОГОВАЯ ПРОВЕРКА · 40 ВОПРОСОВ

Вопросы открытого типа. Сверяйте ответы с учебником. За каждый полностью правильный ответ — 1 балл; 28/40 — зачёт.

1. Объясните основной механизм, рассматриваемый в теме «Микроорганизмы в повседневной жизни».
2. Объясните основной механизм, рассматриваемый в теме «Микроорганизмы в промышленности и медицине».
3. Объясните основной механизм, рассматриваемый в теме «Полимеразная цепная реакция».
4. Объясните основной механизм, рассматриваемый в теме «Применение ПЦР».
5. Объясните основной механизм, рассматриваемый в теме «Генная инженерия».
6. Объясните основной механизм, рассматриваемый в теме «Применение генной инженерии».
7. Объясните основной механизм, рассматриваемый в теме «Этика применения ГМО».
8. Сравните два явления или компонента, упомянутых в теме «Микроорганизмы в повседневной жизни».
9. Сравните два явления или компонента, упомянутых в теме «Микроорганизмы в промышленности и медицине».
10. Сравните два явления или компонента, упомянутых в теме «Полимеразная цепная реакция».
11. Сравните два явления или компонента, упомянутых в теме «Применение ПЦР».
12. Сравните два явления или компонента, упомянутых в теме «Генная инженерия».
13. Сравните два явления или компонента, упомянутых в теме «Применение генной инженерии».
14. Сравните два явления или компонента, упомянутых в теме «Этика применения ГМО».
15. Как изменение условий повлияет на процесс из темы «Микроорганизмы в повседневной жизни»?
16. Как изменение условий повлияет на процесс из темы «Микроорганизмы в промышленности и медицине»?
17. Как изменение условий повлияет на процесс из темы «Полимеразная цепная реакция»?
18. Как изменение условий повлияет на процесс из темы «Применение ПЦР»?
19. Как изменение условий повлияет на процесс из темы «Генная инженерия»?
20. Как изменение условий повлияет на процесс из темы «Применение генной инженерии»?
21. Как изменение условий повлияет на процесс из темы «Этика применения ГМО»?
22. Приведите пример практического применения знаний по теме «Микроорганизмы в повседневной жизни».
23. Приведите пример практического применения знаний по теме «Микроорганизмы в промышленности и медицине».
24. Приведите пример практического применения знаний по теме «Полимеразная цепная реакция».
25. Приведите пример практического применения знаний по теме «Применение ПЦР».
26. Приведите пример практического применения знаний по теме «Генная инженерия».
27. Приведите пример практического применения знаний по теме «Применение генной инженерии».
28. Приведите пример практического применения знаний по теме «Этика применения ГМО».
29. Назовите возможную ошибку в понимании темы «Микроорганизмы в повседневной жизни» и исправьте её.
30. Назовите возможную ошибку в понимании темы «Микроорганизмы в промышленности и медицине» и исправьте её.
31. Назовите возможную ошибку в понимании темы «Полимеразная цепная реакция» и исправьте её.
32. Назовите возможную ошибку в понимании темы «Применение ПЦР» и исправьте её.
33. Назовите возможную ошибку в понимании темы «Генная инженерия» и исправьте её.
34. Назовите возможную ошибку в понимании темы «Применение генной инженерии» и исправьте её.
35. Назовите возможную ошибку в понимании темы «Этика применения ГМО» и исправьте её.
36. Объясните основной механизм, рассматриваемый в теме «Микроорганизмы в повседневной жизни».
37. Объясните основной механизм, рассматриваемый в теме «Микроорганизмы в промышленности и медицине».
38. Объясните основной механизм, рассматриваемый в теме «Полимеразная цепная реакция».

39. Объясните основной механизм, рассматриваемый в теме «Применение ПЦР».

40. Объясните основной механизм, рассматриваемый в теме «Генная инженерия».

ОРИГИНАЛЬНАЯ ГЛАВА С ИЛЛЮСТРАЦИЯМИ

Следующие страницы воспроизведены из загруженного пользователем учебника без изменения содержания.

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CHAPTER 13.0

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CHAPTER 13

BIOTECHNOLOGY

1. Microorganisms in daily life
2. Microorganisms in industry and medicine
3. Polymerase chain reaction
4. Application of PCR
5. Genetic engineering
6. Application of genetic engineering
7. Ethics of GMO application



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13.1 Microorganisms in daily life

You will:

- discuss the advantages and disadvantages of living organisms used in biotechnology.

STQ

Can microbes be useful?

Key terms

- Biofertilizer - is a fertilizer which contains living microorganisms which help plant growth,
- Acidification - is the buildup of hydrogen cations, also called protons, reducing the pH;
- Aflatoxins - are poisonous carcinogens that are produced by certain molds (Aspergillus flavus and Aspergillus parasiticus) which grow in soil, decaying vegetation, hay, and grains;
- Nitrogen fixation - is a process by which nitrogen in the Earth’s atmosphere is converted into ammonia (NH_3) or other molecules available to living organisms.

Research time

Probiotics are the important components of some dairy products. What are probiotics? Which dairy products contain probiotics? How do probiotics affect human health? Do a research and write a summary of one page about it.

Text

Biotechnology is the use of living organisms, such as bacteria, fungi and algae to supply us with useful substances or processes.

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Agriculture

Use of microorganisms in agriculture is concerned with the relationship between microbes and crop plants. Purposes of these relationship are to improve yields, fight against plant diseases and improve resistance to abiotic factors.

One of the examples of microorganism application in agriculture is biofertilizers. Biofertilizers consist of living microorganisms, and they are added to the soil to increase the plant growth. They provide plants with increased amount of nutrients. Commonly used biofertilizers include phosphate-solubilizing bacteria, which solubilize bound phosphates and make them available for plants. Mycorrhizal fungi associated with plant roots, are often important for nutrient uptake and plant survival in natural ecosystems. Azospirillum bacteria stimulate plant growth through nitrogen fixation and production of growth substances.

However, there are some negative aspects of using biofertilizers. Extensive and long-term application of biofertilizers may result in accumulation of salts, nutrients, and heavy metals in soil. This causes adverse effects on plant growth, development of organisms in the soil, water quality, and human health. Also, large volumes of biofertilizers are required for land application due to low contents of soil nutrients, in comparison with chemical fertilizers.

Daily life

People have used microorganisms for thousands of years. Yeasts and some bacteria have been used in baking and wine production.

Baking different types of bread needs different ingredients, yeasts and some types of bacteria. For example, salt plays a contributing factor to the leavening of sourdough bread, bacteria will not grow in media with large amounts of salt. But not all bacteria are beneficial for bread production. Some bacteria, like “ropy bread”, cause infections that have negative effects on the bread.

Lactic acid bacteria, yeasts and molds are used in preparation of yoghurt, cheese, sour cream and other dairy products. Acidification is carried out by lactic acid bacteria whereas other bacteria, molds, and yeasts become dominant during ripening and contribute to the development of aroma and texture in dairy products.

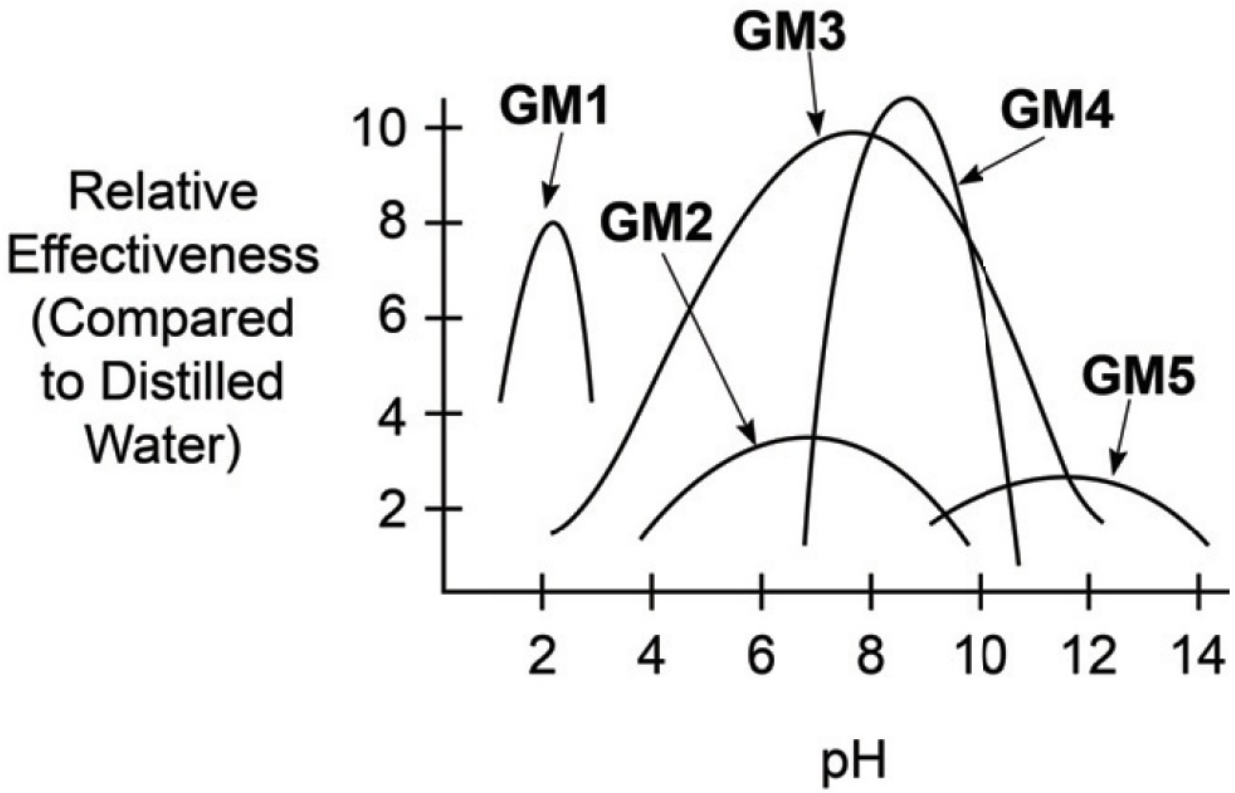
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Fermented dairy products can contain beneficial compounds (vitamins, bioactive compounds), which are produced by the metabolic activity of their microorganisms, but sometimes some microorganisms can also release toxic compounds, such as aflatoxin. Aflatoxin is poisonous carcinogen that is produced by certain molds.

Activity

Enzymes are used as meat tenderizers (processing of meat) in food industry. Some strains of bacteria (GM1-GM5) contain enzymes that act as proteases. Enzyme extracts of bacteria were tested for use as possible meat tenderizers. The extracts were tested over a range of pH values. Analyze the graph below and answer the following questions.



1. Over which pH range does GM3 extract operate at no less than 50% of its maximum effectiveness?
2. Which strain extract is the most effective at a neutral pH?
3. What will happen when an enzyme extract has a relative effectiveness of 1?
4. Which strain extracts are the most effective at acidic pH?
5. Which strain extract is the most effective at a pH of 12?

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Facts

- Genes of Bacillus thuringiensis are used to improve the tolerance of plants against insects. The first insect tolerant genetically modified plant was tobacco, which contained Bt genes from B.thuringiensis.
- Bacteria associated with bread spoilage include Bacillus mesentericus, responsible for “ropy” bread, and it is not particularly toxic. Enzymes secreted by B. mesentericus change the starch inside the loaf into a gummy substance.

Literacy

1. In which fields can microbes be used?
2. What are negative effects of using microbes in daily life?

Terminology

- acidification - қышқылдандыру / подкисление;
- adverse - жағымсыз / неблагоприятный;
- biofertilizer - биотыңайтқыш / биоудобрение;
- gummy - шайырлы / смолистый;
- leavening - ашытқылау / заквашивание;
- nitrogen fixation - азот жинақтау / азотфиксация;
- ripening - пісу / созревание;
- sourdough bread - ашытылған нан / хлеб на закваске;
- to concern - иелену / приобрести;
- to contribute - жүргізу / водить;
- to solubilize - жинау / собирать.

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13.2 Microorganisms in industry and medicine

You will:

- discuss the advantages and disadvantages of living organisms used in biotechnology.

STQ

How are microorganisms used in medicine?

Key terms

- Antibodies - are used by the immune system to destroy pathogens;
- Antimicrobial resistance - is the ability of a microorganism to stop an antimicrobial effect;
- Bioleaching - is the extraction of metals from their ores by microbes;
- Bioreactor - is a device in which biological active substances are produced by iving organisms;
- Growth medium - is a medium on which microbes grow.

Research time

Animals are also used in biotechnology. Do a research about positive and negative aspects of using animals in medicine.

Text

Microorganisms are the main living things which are used in most of the processes of biotechnology. They can be used in industry and medicine.

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Industry

In industry, they are applied in metal bioleaching and production of different metabolites, such as enzymes, amino acids, organic acids, vitamins, insecticides, etc.

Important metals, including iron, copper, zinc are mainly found in the form of metal sulfides, which are insoluble in water. Several types of bacteria are able to oxidise them into metal sulfates, which are soluble in water and can be washed out from rocks. Acidithiobacillus ferrooxidans is a bacterium which converts iron sulfide (FeS2) into iron sulfate (Fe2(SO4)2).

Bioleaching has several advantages to conventional way. It is safe, because it doesn’t produce toxic gas sulfur dioxide (SO2). It can be used to extract metals from ash and industrial wastes. Bioleaching reduces production costs. It helps to get the metal from low grade ore.

However, sometimes bioleaching has negative effects. It is very slow process compared to smelting. Sulfuric acid is produced, which may leak into the ground and water surface, making it acidic, and thus causing environmental damage.

Medicine

Microorganisms in medicine are used to produce antibiotics, vaccines, antibodies, etc. Microorganisms are very small and are easy to grow in laboratory, because the reproduce very quickly. Their DNA can be mutated or genetically engineered, and their genes are quite easy to transfer from one cell to another via plasmids.

The pharmaceuticals produced by microbes should be selectively toxic to the microbe but nontoxic to the host cells. They should remain long enough to act and not to be broken down or excreted prematurely. Microbes should not lead to the development of antimicrobial resistance. They should remain active in tissues and body fluids. Also, they should be reasonably priced.

There are also negative effects of using microorganisms in medicine:

- during preparation of vaccines, laboratory assistant should be careful, because they are pathogens;

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- if the production of microorganisms is disrupted, they can mutate rapidly and cause diseases;
- inactive pathogenic microorganisms can be activated in some cases, so the pathogenic characteristics always must be regulated;
- the lifespan of some pharmaceuticals produced by microbes is very short;
- some of the pathogenic microbes cannot be cultivated in a laboratory.

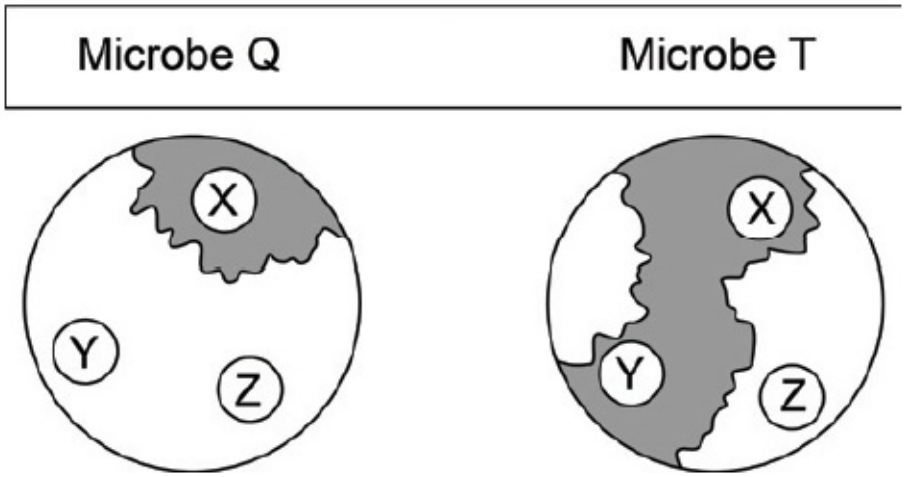
Activity

One scientist identified three bacterial strains: Q, R, and T. Three different growth media were prepared, each containing only 4 amino acids. Growth was determined in these environments. The amino acid content was as follows:

- Medium 1: cysteine, phenylalanine, serine, threonine;
- Medium 2: cysteine, phenylalanine, proline, tryptophan;
- Medium 3: cysteine, proline, threonine, tryptophan.

Results are given in a table below: '+' - presence of growth, '-' - absence of growth.

	Q	*R*	*T*
Medium 1	+	+	+
Medium 2	-	+	+
Medium 3	+	-	-



Scientist further determined that the microbes R was not pathogenic. Three

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potential antibiotics X, Y, and Z were tested against strains Q and T. Below, the effects of these antibiotics on growth after 24 hours of bacterial growth are shown. The painted area corresponds to a good growth of bacteria.
Based on given above data, find whether the following statements are true or false.

1. _____ The cell membrane is present in the microbe R.
2. _____ Threonine is essential for the bacterium Q.
3. _____ Serine is essential for the bacterium Q.
4. _____ Phenylalanine is not essential for the bacterium Q.
5. _____ By observing microbe T, scientist would not find a nuclear membrane.
6. _____ Medium 1 can be used to determine the effectiveness of antibiotics against pathogenic bacteria.
7. _____ Antibiotic Z is equally effective against bacteria Q and T.
8. _____ Antibiotic Y is effective against bacteria T, but not Q.

Facts

- Bioleaching is preferable in extracting gold rather than copper, because the concentration of gold is low in its ore compared to copper.
- Antibodies are used to prepare immunotoxins, which consists of antibody and toxic component. These type of new generation drugs bind to specific antigen of cancer cells, and acts only on it by toxin, not other normal cells.

Literacy

1. Why are microorganisms conevnient to use in industry and medicine?
2. What are negative effects of using microbes in medicine?

Terminology

- antimicrobial resistance - антимикробтық төзімділік / противомикробная устойчивость;
- bioleaching - биосілтісіздендіру / биовыщелачивание;
- conventional - дәстүрлі / традиционный;
- growth medium - қоректік орта / питательная среда;

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- ore - кен / руда;
- prematurely - мезгілсіз / преждевременно;
- smelting - қорыту / плавка;
- to disrupt - бүлдіру / срывать;
- to wash out - жуып шығару / вымыть.

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13.3 Polymerase chain reaction

You will:

- describe the importance of PCR in taxonomy, medicine and forensic science.

STQ

How scientists copy DNA in the lab?

Key terms

- Genetic marker - is a gene or DNA sequence with a known location on a chromosome that can be used to identify individuals or species;
- Restriction enzymes - enzymes that cuts DNA at specific sites.

Research time

Sometimes scientists use method which is called Reverse transcriptase-PCR (RT-PCR). Do a research about it, and write the steps of RT-PCR. Find when scientists use this method.

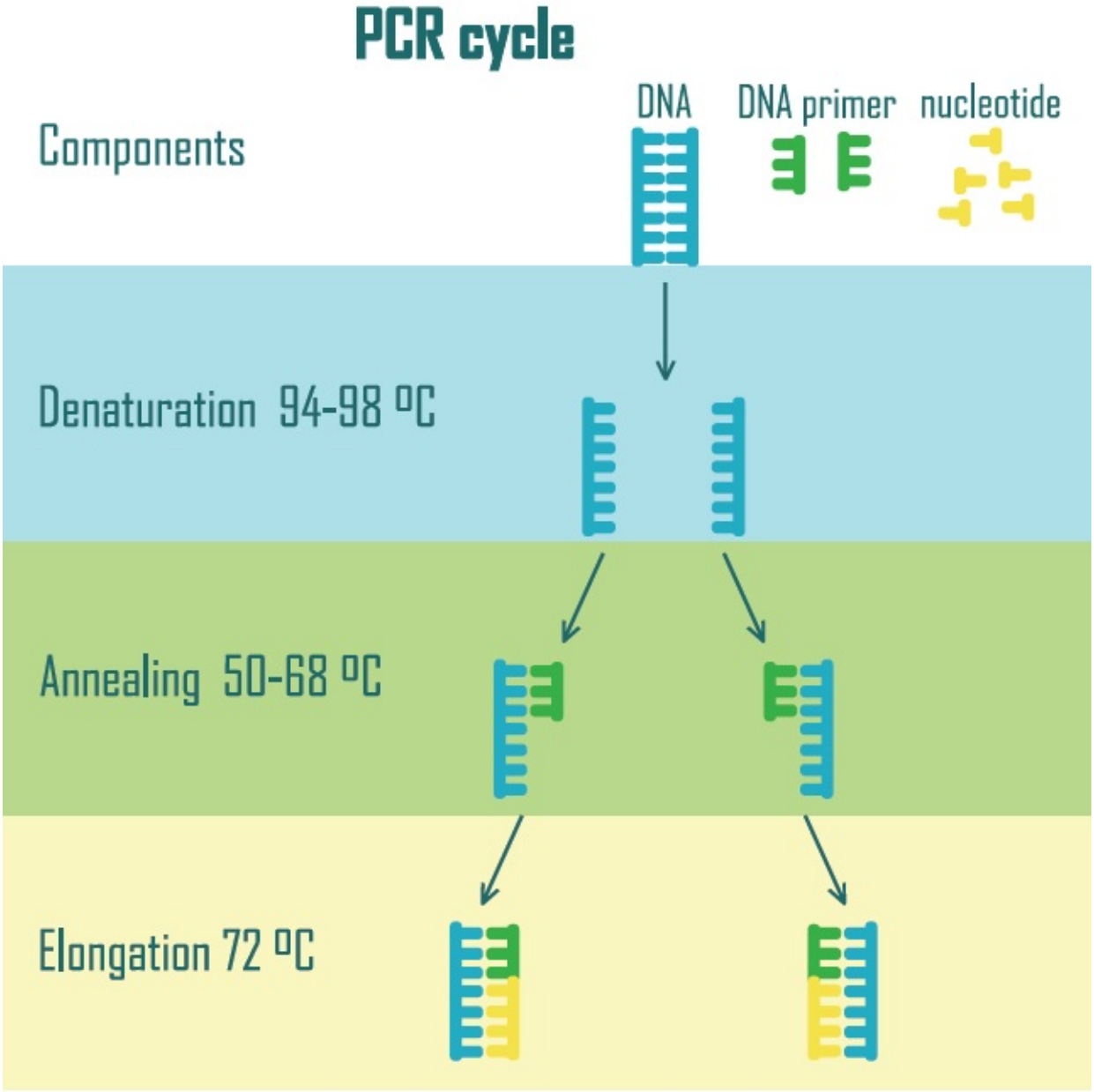
Text

The polymerase chain reaction (PCR) is a method for production of many copies of a particular fragment of DNA. Within a few hours, this technique can make billions of copies of a target DNA segment.
There are machines in laboratories, which perform PCR. The materials needed for PCR are small, thin tubes for reaction, and reaction solution. Reaction solution contains DNA template, DNA primers, free nucleotides, a buffer solution and DNA polymerase (Taq polymerase) enzyme. The main important factor is temperature, that automatically changes by PCR machine.
The PCR includes 3 steps: 1) Denaturation: DNA is denatured at 95OC, and

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double helix is separated, 2) Annealing: after cooling, DNA primer is bind to particular DNA fragment by complementary base pairing, it requires a temperature of about 65OC, 3) Elongation: DNA polymerase adds free nucleotides to synthesize complementary strand. This requires a temperature of around 72OC. First copy of DNA is ready, and above three steps occur again and again.



Stages of PCR

Importance of PCR in forensic science and paternity test

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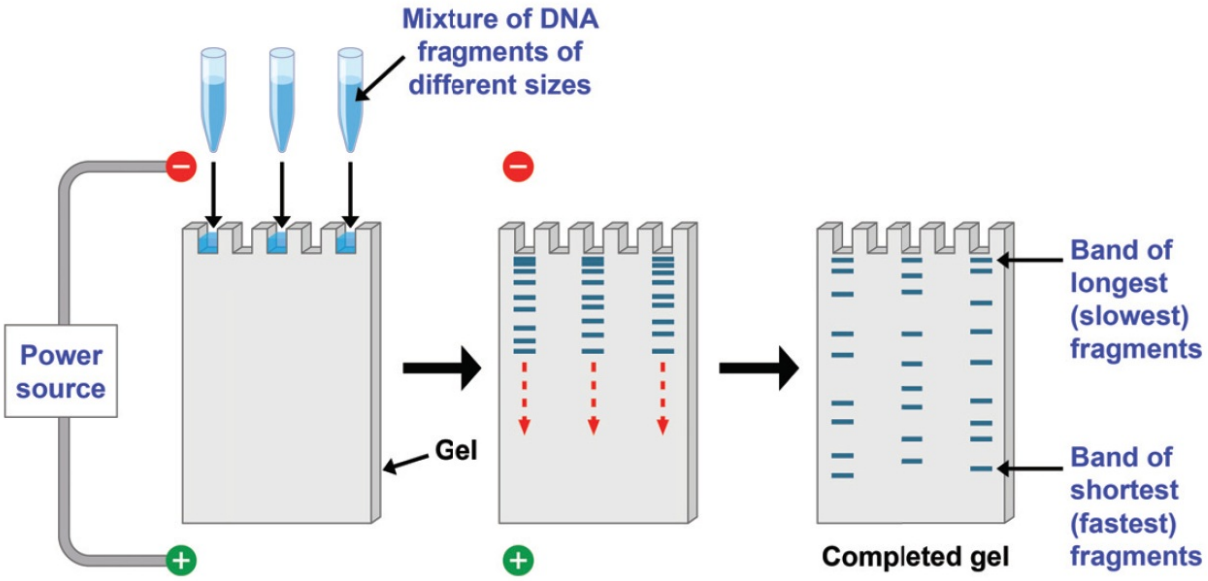
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The DNA sequence of every person is unique (except for identical twins). Genetic markers vary in the population. This difference can be analyzed for a given person to determine that individual's unique set of genetic markers. This method is called DNA profiling. One of these genetic markers are short tandem repeats (STRs). The number of repeats varies from person to person. Individuals will likely have different numbers of repeats at a given DNA locus, they will generate unique DNA profiles. For example, in the image, you can see that AGAT sequence can be repeated differently in individuals.

PCR is used to make copies of particular STRs. In this reaction, PCR uses sets of primers that are labeled with different colored fluorescent tags. The copies of DNA is then cut into fragments by restriction enzymes. The restriction enzymes cut DNA close to the STR regions. The length of the region and the number of repeats, can then be determined and visualized by electrophoresis.

Gel electrophoresis is a technique used to separate different sized molecules of DNA by electric field. The fragments of DNA move through the gel towards the anode. Short fragments move far away from the cathode than long fragments. After electrophoresis, the gel with DNA fragments is placed on an X-ray film. The radiation emitted by fluorescent tags make the film go dark. The pattern of dark stripes (bands) on the film matches the positions that the DNA fragments reached on a gel.

Analysis of genetic markers of people helps to find suspect in a crime, and father in paternity test.



Gel electrophoresis of DNA

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Activity

1. Using the following primer: 5’ - CTGGA, locate where it binds to the DNA template.

Primer: 5′- -3′
DNA template: 3′-TAGCTATGCGGACCTCATGCATTAGAGTAG-5′

2. Fill the table to illustrate the steps of the PCR and describe what happens on each step. Use the following words:

template	primers	Taq polymerase	denaturation	elongation	annealing

Name of step			
Temperature			
Description			
Illustration			

Facts

- Taq polymerase was the first heat-stable DNA polymerase to be used in PCR. It was isolated from the thermophilic bacterium, Thermus aquaticus, which is found in hot springs in Yellowstone Park in the USA.

Literacy

1. How many molecules of DNA are produced after 8 cycles of PCR?
2. Why DNA molecule moves towards cathode during gel electrophoresis?
3. Why fluorescent tags are used during gel electrophoresis?

Terminology

- band - сызық / полоска;
- crime scene - қылмыс орны / место преступления;
- fluorescent tags - флуоресцентті таңбалар / флуоресцентные метки;

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- individual - дарақ / особь;
- paternity test - әкелікті тексеру / тест на отцовство;
- polymerase chain reaction - полимеразалық тізбекті реакция / полимеразная цепная реакция;
- primer - праймер;
- short tandem repeats - қысқа тандемдық қайталанымдар / короткие тандемные повторы;
- to label - белгілеу / метить;
- to vary - ауысу / меняться;

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13.4 Application of PCR

You will:

- describe the importance of PCR in taxonomy, medicine and forensic science.

STQ

Why is PCR needed?

Key terms

- Genetic marker - is a gene or DNA sequence with a known location on a chromosome that can be used to identify individuals or species;
- DNA cloning - enzymes that cut DNA at specific sites.

Research time

There are many ways of using PCR in molecular biology and genetic engineering. PCR is also used in mutagenesis, creating mutations in a gene and obtaining many copies of that gene. Make a research how PCR is used in mutagenesis.

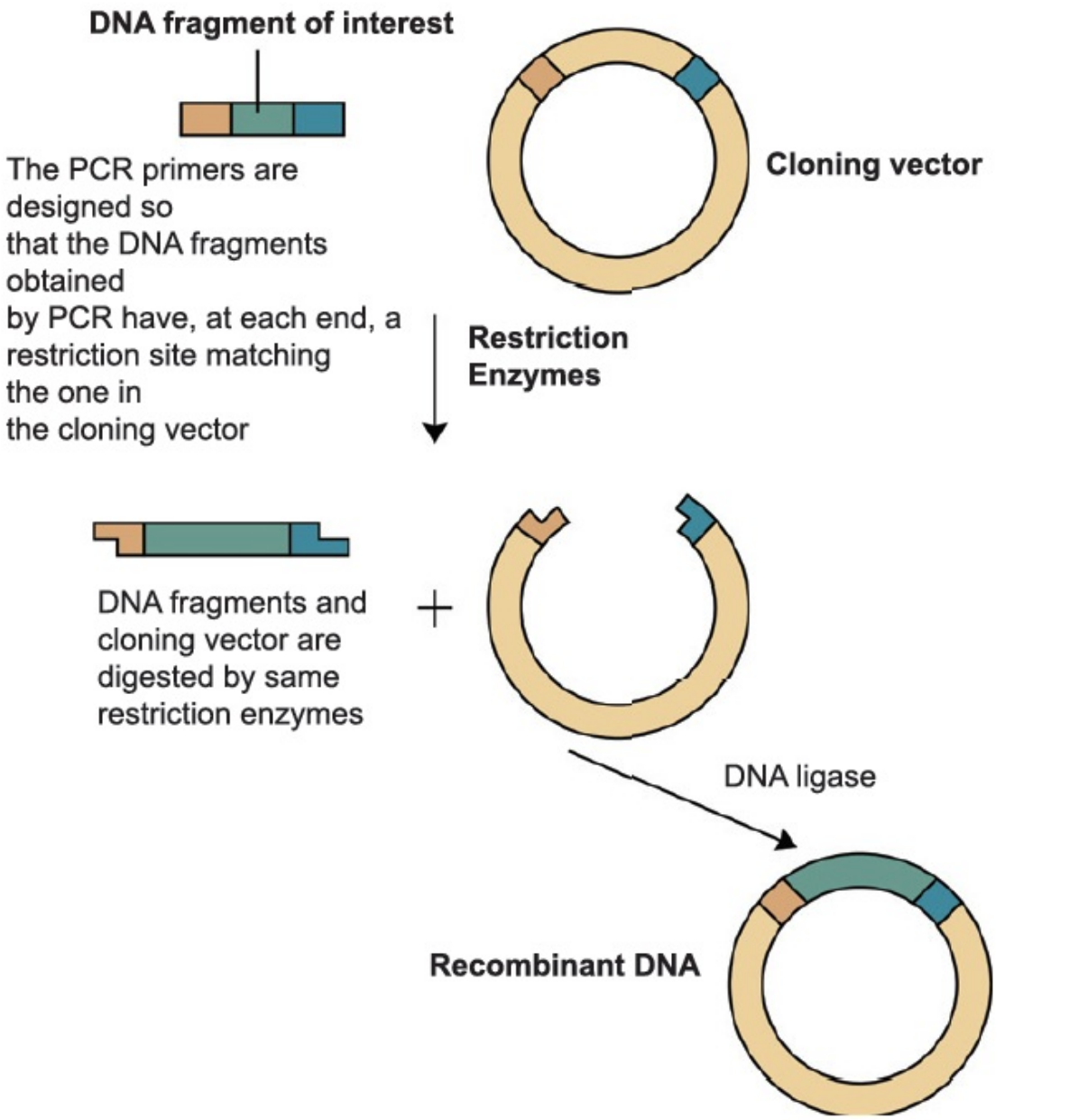
Text

PCR in cloning genes

PCR is used to supply the specific DNA fragment for cloning. PCR primers are designed to have a restriction site at each end of the DNA fragment obtained by PCR. It matches the site in the cloning vector. Then the fragment and vector are cut, hybridized, and ligated together.

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Use of a restriction enzyme and PCR in gene cloning

PCR in diseases diagnosis

One important use of PCR is the identification of human genes whose mutation or influence plays role in genetic diseases and 'non-genetic' diseases, such as arthritis and AIDS. These discoveries may lead to ways of diagnosing, treating, and even preventing such conditions.
Medical scientists can diagnose hundreds of human genetic disorders by using

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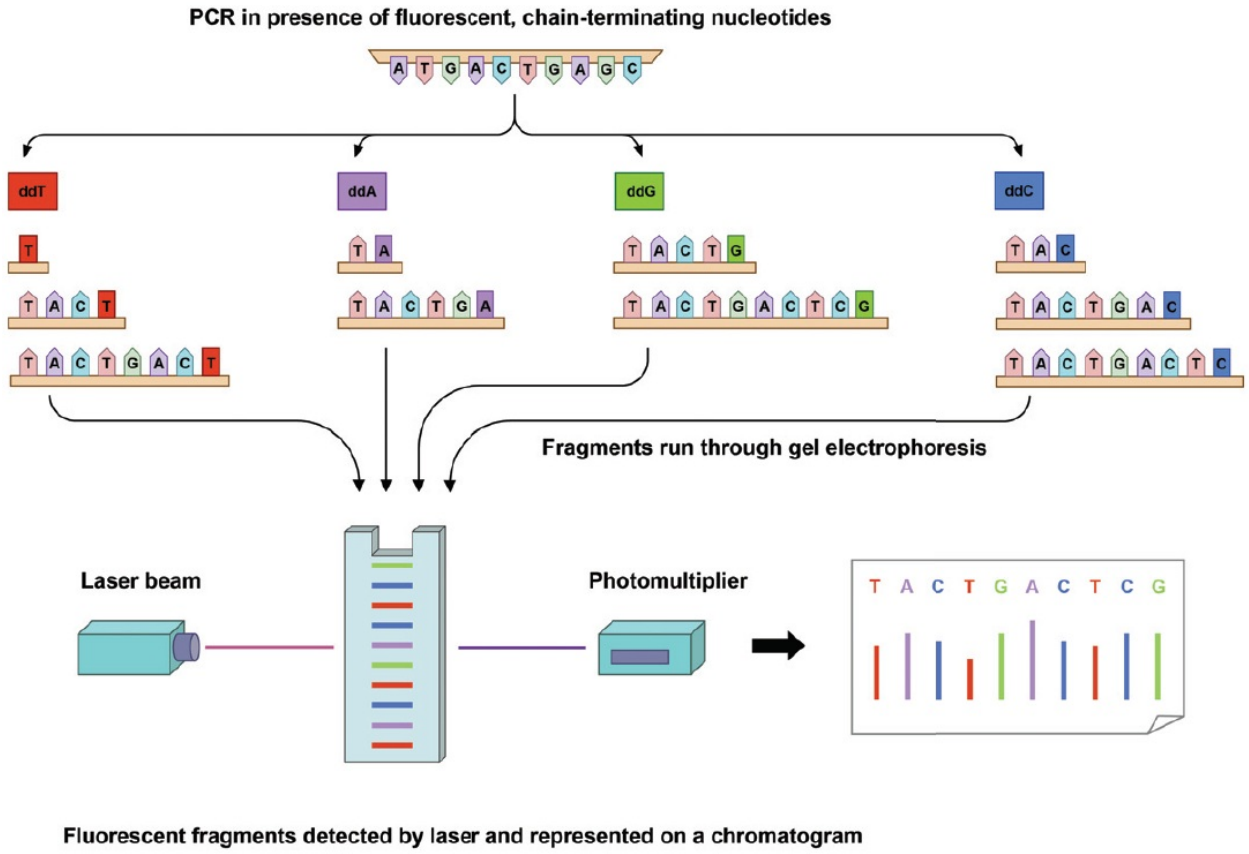
PCR. The primers that target the genes associated with these disorders are used. The DNA product after PCR is then used to find the presence or absence of the disease-causing mutation.
By “personalized medicine”, genetic makeup of each individual can inform them about diseases or conditions for which they are especially at risk and help them make treatment choices.

PCR in sequencing DNA molecule

Sequencing of DNA molecule is determining nucleotide order of DNA nucleotides. This process can be performed by PCR. The difference is that some of the primers are modified and electrophoresis is performed in capillary tube. Modified primers are tagged by fluorescent dye (4 different colours) and doubly deoxidised nucleotides (ddNTP). These primers stop the PCR when, by chance, they join to the growing DNA, because they lack a 3’-OH group, the site for attachment of the next nucleotide.
The result is many copies of DNA with different lengths, and ending is tagged by colour. There are as many different lengths as the number of bases in gene. The mixture with different lengths of DNA is then pulled towards cathode, along capillary tube by electrophoresis. At the end there is a laser light, which shines and colour is recorded automatically. Computer records sequence of colours, which represent particular bases that pass through the laser. Short ones reach the laser first, so these bases were closer to the beginning of the DNA. The sequence of bases passing through the laser determines the sequence of bases in the DNA.

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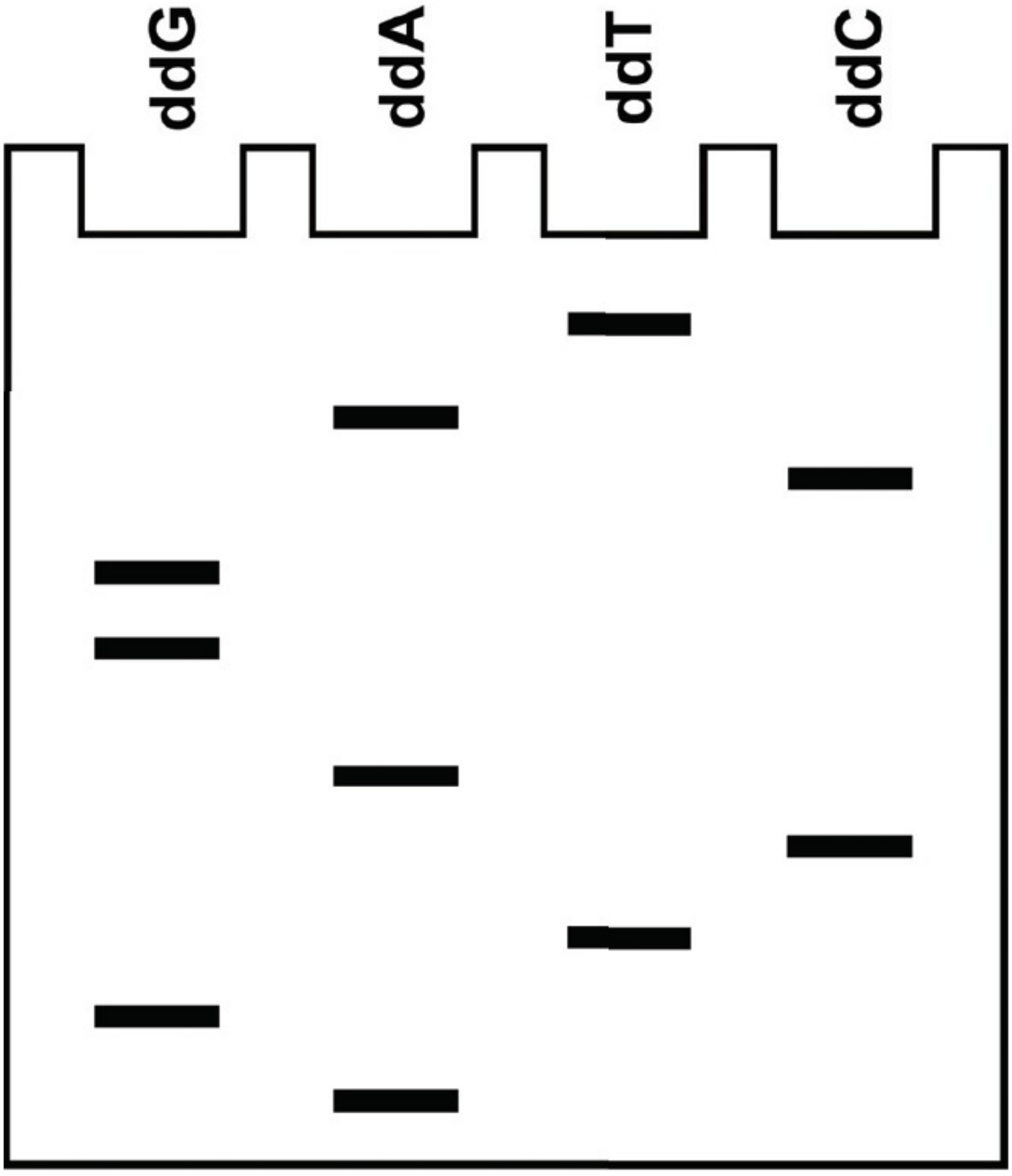
Process of PCR in sequencing DNA molecule

Activity

Imagine that you are working in a laboratory as a lab assistant. You have a small fragment of DNA that you are going to sequence. You perform PCR using modified primers. Below is the resulting gel. What is the sequence of the original DNA? Explain your answer.

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Facts

- Kary Mullis earned Nobel Prize in 1993, for recognition of his invention of the PCR technique.

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- DNA can be extracted from almost anything that has come from a person’s body – the root of a hair, a tiny spot of blood or semen at a crime scene, or saliva where someone has drunk from a cup.

Literacy

1. What is the function of PCR?
2. Where PCR is used?
3. What is sequencing?

Terminology

- at risk - қауіпте / в опасности;
- crime scene - қылмыс орны / место преступления;
- electrophoresis - электрофорез;
- genetic makeup - генетикалық құрамы / генетический состав;
- invention - өнертабыс / изобретение;
- sequencing - секвенирлеу / секвенировать;
- tagged - таңбаланған / меченый;
- to assemble - жинау / собрать;
- to determine - анықтау / определить;
- to record - тіркеу / регистрировать;
- to target - бағыттау / направлять.

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13.5 Genetic engineering

You will:

- explain the stages of genetic engineering manipulations.

STQ

How to insert genes into an organism?

Key terms

- Sticky ends - short lengths of unpaired bases;
- Vector - delivers recombinant DNA to the cells of the organism.

Research time

Make a research and write at least three advantages and disadvantages of genetic engineering.

Text

Genetic engineering involves transfer of a gene from one organism to another. Transferred gene should be expressed (producing protein) in new host organism. The new DNA molecule made from two different organisms is called recombinant DNA. The organism which now expresses the new gene or genes is known as a transgenic organism or a genetically modified organism (GMO).

The steps of genetic engineering manipulations

The are many ways of producing GMO, but following steps are general principles of genetic engineering.

Production of human insulin:

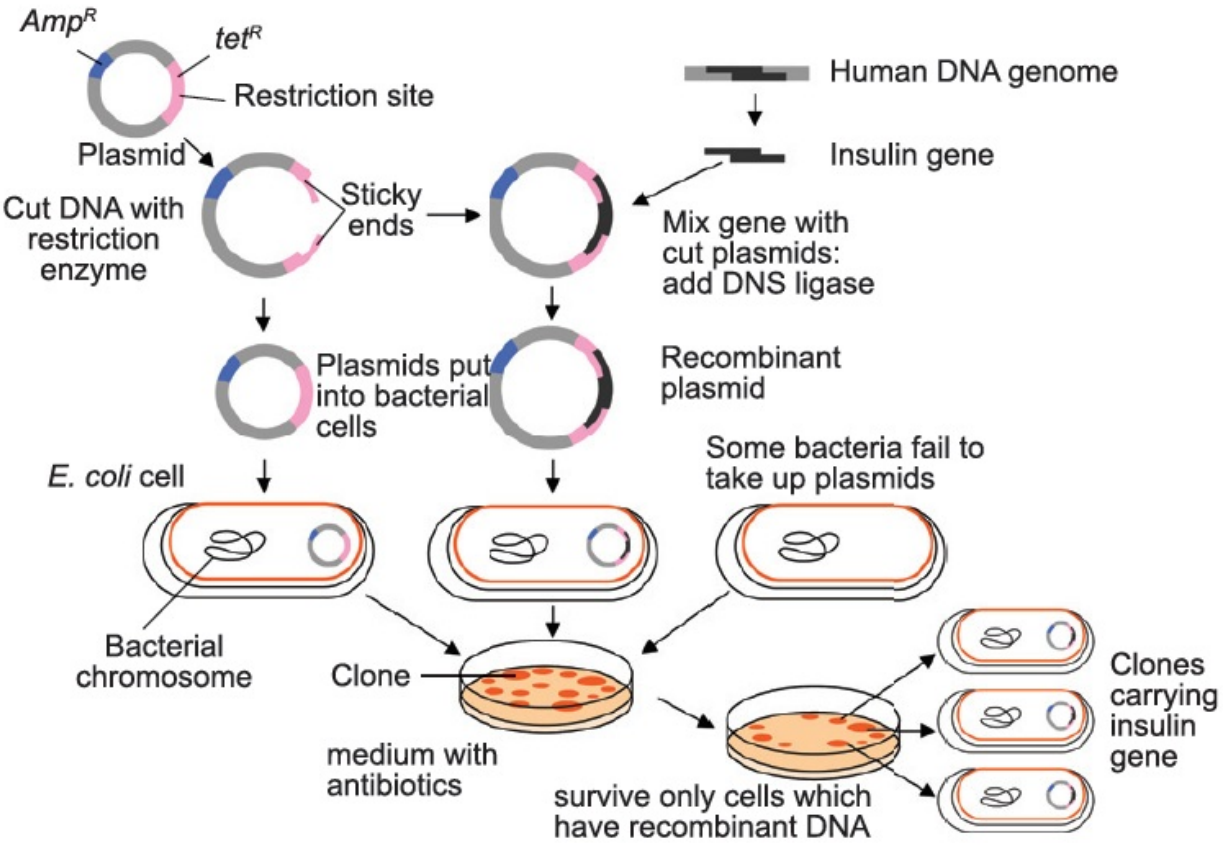
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1. Human insulin gene is isolated from a chromosome by restriction enzymes, for example, EcoRI. They produce sticky ends. Sticky ends can easily form hydrogen bonds with complementary sequences of bases on another DNA cut with the same restriction enzyme;
2. Plasmid DNA (vector) is cut with the same restriction enzyme, and sticky ends are produced. pBR322 is a vector, which contains origin of replication (ori) where replication starts, antibiotics resistance genes, and restriction sites for restriction enzymes, such as EcoRI, BamHI, and PstI;
3. The insulin and plasmid DNA are joined by sticky ends, and enzyme ligase joins the DNA backbone. The new molecule is now called recombinant DNA;
4. The recombinant DNA is introduced into bacteria;
5. The bacteria which take up recombinant DNA are identified. For this purpose, they are grown on a medium which contains ampicillin or tetracycline antibiotics. Bacteria which take up recombinant DNA survive, because they have antibiotics resistance gene;
6. The survived bacterial cells are cloned, and grown at an industrial scale in large amounts;
7. Human insulin is extracted from the bacteria and purified.

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The steps of introducing human insulin gene into bacterial cells

Genetic engineering is used in many ways in our life, these are some fields of its application:

- medicine - production of pharmaceutical products, such as hormones, vaccines and antibiotics;
- agriculture - GM plants and animals;
- environmental cleanup - biofuels and biodegradation of wastes;
- industry - production of enzymes, washing powders, dietary products, etc.

Activity

Work in pairs to answer following questions.
A eukaryotic gene, which is responsible for growth hormone, has “sticky ends” produced by the restriction endonuclease BamHI.

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This gene is added to a mixture containing BamHI and a bacterial plasmid pBR322. This mixture is incubated for several hours, exposed to DNA ligase, and then added to nutrient medium where this bacteria grow. The bacteria are allowed to grow overnight and they produce colonies. Samples of these colonies are then grown in four different media:
A) nutrient medium + ampicillin
B) nutrient medium + tetracycline
C) nutrient medium + ampicillin and tetracycline
D) nutrient medium without antibiotics

1. The gene will be inserted into which part of plasmid? _____
2. Bacteria that contain the plasmid, but not the eukaryotic gene, would grow in _____
3. Bacteria containing a plasmid into which the eukaryotic gene has integrated would grow in _____
4. Bacteria that do not take up any plasmids would grow on which media?

Facts

- Poplar tree is the product of genetic engineering that can absorb polluted water through their roots and clean it before the water is released back into the air.

Literacy

1. List three advantages of genetic engineering methods compared to selective breeding methods.
2. What is a vector and why it is needed?
3. Why is antibiotic resistance gene is necessary in a plasmid?

Terminology

- biodegradation - биологиялық ыдырау / биологический распад;
- industrial scale - өндірістік мөлшер / промышленный масштаб;
- medium - орта / среда;
- poplar tree - терек / тополь;
- sticky ends - жабысқақ ұштар / липкие концы;

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- to clone - клондау / клонировать;
- to express - көрсету / выражать;
- to take up - орын алу / занять;
- transgenic organism - трансгенді организм / трансгенный организм;
- vector - вектор.

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13.6 Application of genetic engineering

You will:

- explain the stages of genetic engineering manipulations.

STQ

Where genetic engineered organisms can be used?

Key terms

- Cephalosporins - are a class of β-lactam antibiotics originally derived from the fungus Acremonium, which was previously known as “Cephalosporium”;
- Chlorinated hydrocarbons - are very potent nerve toxins to insects, and their initial use led to significant improvements in insect control;
- Growth factors - compounds that are capable of stimulation of growth, division and differentiation of cells;
- Vaccine - immunobiological drug, which is used to produce immunity against infectious diseases.

Research time

Do the research to the given organisms and explain their role in the genetic engineering. How scientists manipulate their genes, is it harmful to these organisms? Explain your answer with evidence.

- Escherichia coli
- Drosophila melanogaster
- Arabidopsis thaliana
- Zebrafish

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- Tobacco mosaic virus

Text

Medicine

Pharmaceutical proteins produced by genetic engineering are insulin, growth hormone, factor VIII - blood clotting factor, growth factors, etc. Growth factors are used to treat ulcers in stomach and scars on skin.
Transgenic plants have been engineered to produce vaccines that can be taken effectively by mouth. They include vaccines against rabies and cholera. Banana has been used, which is cheap and widespread.
The development of recombinant DNA techniques and their application to antibiotic producing microorganisms has allowed yield increments. The design of biosynthetic pathways gives rise to new antibiotics. For example, genetically engineered fungus Cephalosporium acremonium, produced more cephalosporin (antibiotics), and produced new penicillin G.

Agriculture

Desirable traits in agricultural crops, such as enhancement of shelf life of fruits and vegetables, delayed ripening and resistance to spoilage, insects and disease, as well as abiotic factors are interest of genetic engineering. For example, shelf life of transgenic tomato is longer than non-transgenic tomato.
The goals of creating a transgenic animal are milk yield, to make a sheep with better quality wool, a domestic animal with leaner meat, or a cow that will mature in a shorter time. Transgenic salmon is engineered, and it is bigger than non-transgenic salmon.

Environmental cleanup

Pollution of environment is one of the major problems nowadays. Genetically engineered microorganisms can be used in treating polluted water and soil. They degrade oil products, chlorinated hydrocarbons and other harmful compounds, because they have introduced genes responsible for degrading harmful compounds.
Ethanol (alcohol), produced from fermented sugar, could replace, or at least supplement, petrol. Ethanol produces less pollution than petrol. Fatty acids and

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glucose can be converted to ethanol by anaerobic respiration. There are some GM microalgae that produce more fatty acids than non-GM microalgae.

Industry

Fungi (like Aspergillus) or bacteria (like Bacillus) are common organisms used to produce the enzymes. To 'improve' the action and to enhance production of the enzymes coded by the genes, new genes can be introduced into the microbes. Enzymes produced by these microbes are used to clarify juices, enhance flavour, maximize juice extraction, make cheese, produce lactose-free milk in food industry.
Enzymes are used in preparing washing powders. They contain mainly proteases and lipases, to remove egg, meat, blood, fat stains. They save energy, because there is no need for boiling water.

Activity

Divide into groups of 4 members. Select roles and read scenario. Present a family debate.

Roles	*Characteristics*
Mrs Kuralay	genetic engineer
Husband	likes foreign holidays and fast cars
Daughter	vegetarian and member of a local animal rights group
Son	very religious and has strong views on "man playing God"

Scenario: Mrs Kuralay has managed to transfer some cattle growth hormone genes into a chicken. She tells her family that modified chickens will grow even faster. However, fast-growing chickens may have trouble standing, suffer from weak hearts and chronic pain. Chickens can be slaughtered sooner and therefore farmers will kill more chickens per year and make more money. She is going to patent her invention and hopes to make a lot of money from it.

Facts

- Golden rice is the product of genetic engineering which contains beta-carotene 20 times higher than in normal rice and it prevents the vitamin A

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deficiency.

Literacy

1. What characteristics of microorganisms are used in treating polluted ecosystems?
2. Why you should wear gloves while using washing powders which contain enzymes?
3. Explain why lactose-free milk is produced.

Terminology

- cholera - тырысқақ / холера;
- counterpart - көшірме / копия;
- enhancement - күшейту / усиление;
- growth factors - өсу факторлары / факторы роста;
- increment - өсім / прирост;
- leaner - жұқа / тощий;
- microalgae - микробалдырлар / микроводоросли;
- rabies - құтыру / бешенство;
- ripening - пісу / созревание;
- ulcer - жара / язва.

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13.7 Ethics of GMO application

You will:

- discuss ethical issues of application of genetically modified organisms.

STQ

What are disadvantages of GMO?

Key terms

- Crop plants - are plants that are grown by humans for food and other resources;
- Ethical question - involve consideration of conflicting moral choices and dilemmas;
- Organic certification - is a certification process for producers of organic food and other organic agricultural products;
- Transgenic organism - an organism which contains DNA form another organism, and also called genetically modified organism.

Research time

What potential dangers happen if genetically modified microorganisms contact with the outside environment? Make a research and make a mind map.

Text

Genetically modified (GM) organisms are used in some countries, and mainly crop plants rather than animals. How have GM organisms affected our society? Can we see any improvements in our lifestyle and health? Are there any damage and harm from using GMO?

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Benefits of GMO

In the past, the organisms with desirable traits were obtained by selective breeding methods, nowadays we can create GM organisms in a short period of time. By the help of creating GMO, it is possible to improve nutritional content of food, for example, vitamin A rich rice, increase the yields of crops and domestic animals, produce different pharmaceuticals, such as vaccines, antibiotics, hormones. These pharmaceuticals can be produced by GM plants, and used effectively by eating these plants.
Enzymes are made for use in food industry, but obtaining them by traditional methods are ineffective. Rennin is an enzyme which is used in cheese production, traditionally it was obtained from cow calves and it was to obtain enough enzyme to satisfy demand, but now it is produced by GM microbes.

Possible risks of GMO

Generally GMO are seen unnatural and unsafe compared to organisms produced by selective breeding methods. Concerns about GM crops are the modified crop plants may become agricultural weeds or invade natural habitats, the introduced gene(s) may be transferred by pollen to wild relatives and unmodified plants growing on a farm with organic certification, the modified plants may be a direct hazard to humans, domestic animals, by being toxic or producing allergies, GM seeds are expensive, and growers mostly need to buy seeds each season.
One of the worries is that the vectors that carry genes, mainly contain antibiotic resistant genes. if this antibiotic resistance gene is transferred to harmful microbes in our body, they would obtain high resistance to antibiotics.

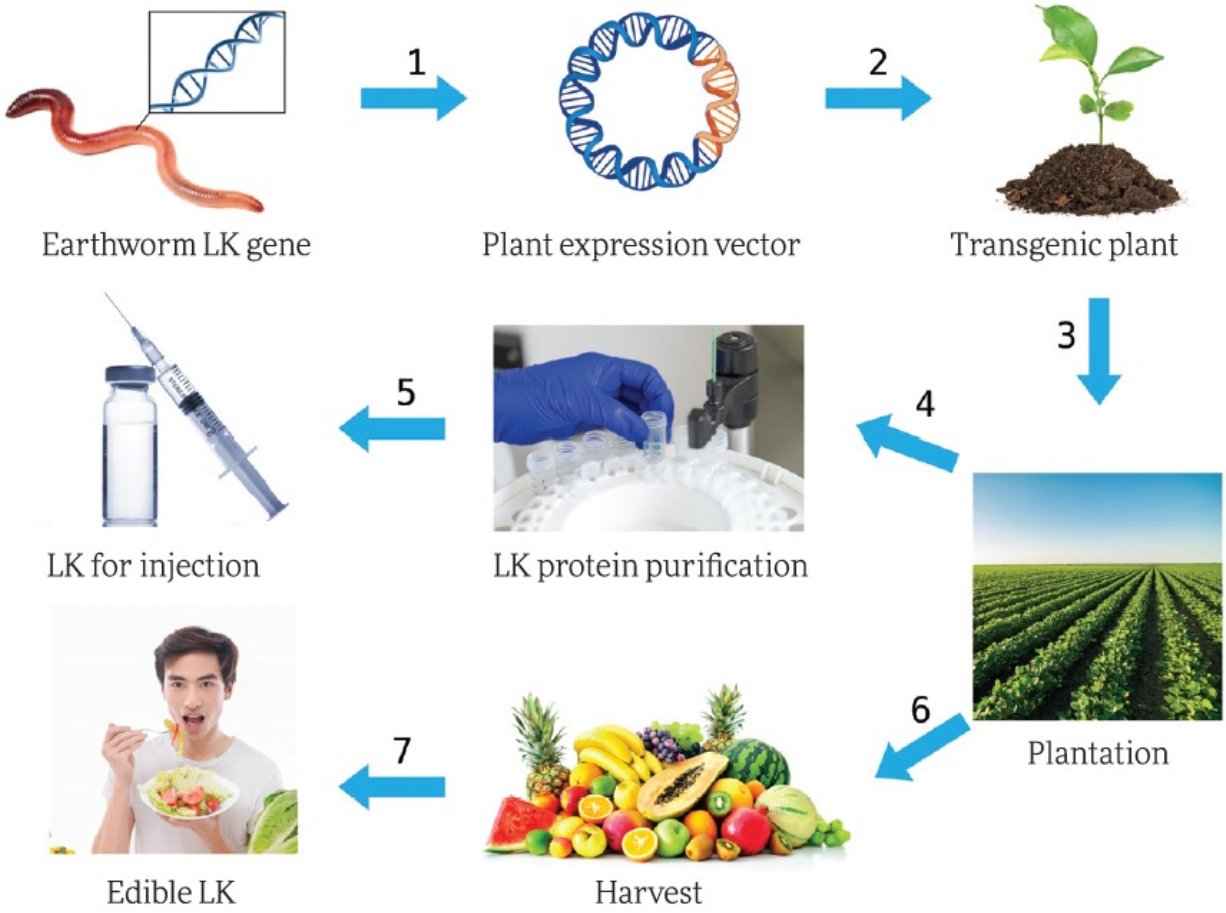
Ethical questions

There is a debate between two groups: 1) some people consider GMO as a solution to food shortage, the scarcity of environmental resources and weeds and pests infestations; and 2) independent scientists, environmentalists, farmers and consumers who warn that GM food introduces new risks to food security, the environment and human health.
There are different levels of ethics for different GMO. For example, the pain of an animal is minimal and potential benefit is high, which makes ethical question an easy one, in case of studying diseases in GM animals.
For now, it is hoped that people will become more educated on the ethical

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concerns about GMO, which will contribute further research and accountability in this field.



Activity

Discuss with your friends following questions by Think-Pair-Share method:

- In what circumstances use of genetic engineering is acceptable?
- Is it acceptable to modify an organism that produces useful product?
- Is it acceptable to engineer an animal to study diseases?
- Is it acceptable to modify crops to improve yields? And in what cases?
- Should products containing GMO be labeled in markets?
- Should we accept use of GMO in our country?

Facts

- In 2009, animals became factories for pharmaceuticals. Genetically

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modified animals produce various drugs which are used in medicine. For example, human alpha-antitrypsin is produced by sheep – this is used to treat people with emphysema
- The first genetically modified crop approved for sale in the U.S., in 1994, was the FlavrSavr tomato. It had a longer shelf life, because it took longer to soften after ripening

Literacy

1. What is GMO?
2. Are GMO bad or good? Explain your answer.
3. Where GMO are used?

Terminology

- accountability - жауапкершілік / ответственность;
- calve - бұзау / теленок;
- demand - қажеттілік / нужда;
- desirable - қажет / желательный;
- food shortage - тағам жетіспеушілігі / дефицит еды;
- harvest - өнім / урожай;
- infestation - зақым / заражение;
- nutritional content - қоректік құрамы / питательное содержание;
- pharmaceuticals - фармацевтикалық препараттар / фармацевтические препараты;
- purification - тазарту / очистка;
- to invade - басып кіру / вторгаться.

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Problems

Test questions with one right answer

1. What are stages of PCR?

A) replication-transcription-translation

B) initiation-elongation-termination

C) denaturation-annealing-elongation

D) prophase-metaphase-anaphase-telophase

E) denaturation-renaturation

2. What is biotechnology?

A) using living organisms to supply us with useful substances

B) using robots in biology

C) using new technology to save the nature

D) creating robots similar to living organisms

E) using smartphones on biology lessons

3. What does EcoRI produce?

A) insulin

B) plasmid

C) vector

D) sticky ends

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E) ligase

4. Extraction of metals from their ores by microbes:

A) bioremediation

B) bioleaching

C) acidification

D) annealing

E) restriction

Test questions with several (max 3) right answers

1. Which living organisms are used in daily life?

A) yeasts

B) Azospirillum bacteria

C) Lactic acid bacteria

D) Molds

E) Acidithiobacillus ferrooxidans

F) Sulfuric acid bacteria

G) E. coli

H) Mycoplasma

2. Materials needed for PCR:

A) DNA ligase

B) H2O

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C) Oxygen gas

D) DNA polymerase

E) bacteria

F) DNA primers

G) restriction enzymes

H) free nucleotides

3. Materials needed for insulin production:

A) DNA primers

B) restriction enzymes

C) ligase

D) plasmid

E) DNA polymerase

F) 95OC temperature

G) buffer solution

H) free nucleotides

Matching questions

1. Match these processes with temperatures they require:

1. Denaturation
2. Annealing
3. Elongation

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A) 36-38C

B) 72C

C) 50-68C

D) 102-104C

E) 94-98C

F) 44-45C

2. Match these organisms with the fields they are used in:

1. Acidithiobacillus ferrooxidans

2. Yeasts

3. Azospirillum bacteria

A) diary products production

B) bread production

C) stimulate plant growth

D) production of aflatoxins

E) bioleaching

F) insulin production

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ANSWERS

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ANSWERS

1. MOLECULAR BIOLOGY AND BIOCHEMISTRY	1	2	3	4
Answers for the tests with one correct answer	A	D	B	C
Answers for the tests with several correct answers	A, C, E	C, D, H	E, F, H	
Matching questions	1-D, 2-E, 3-B	1-B, 2-E, 3-D		

2. CELL BIOLOGY	1	2	3	4
Answers for the tests with one correct answer	B	C	D	D
Answers for the tests with several correct answers	B, E, F	A, D, F	B, C, F	
Matching questions	1-B, 2-F, 3-C	1-E, 2-A, 3-B		

3. NUTRITION	1	2	3	4
Answers for the tests with one correct answer	D	D	D	A
Answers for the tests with several correct answers	A,E,G	A,D,H	C,F,H	
Matching questions	1-D, 2-F, 3-A	1-A, 2-C, 3-E		

4. TRANSPORT	1	2	3	4
Answers for the tests with one correct answer	C	E	B	C
Answers for the tests with several correct answers	C, F	A, E	A, D, G	
Matching questions	1-D, 2-B, 3-E	1-F, 2-C, 3-D		

5. RESPIRATION	1	2	3	4
Answers for the tests with one correct answer	C	C	E	A
Answers for the tests with several correct answers	C, F, G	A, E, G	A, D, G	
Matching questions	1-E, 2-A, 3-C	1-E, 2-C, 3-F		

6. EXCRETION	1	2	3	4
Answers for the tests with one correct answer	A	B	E	D
Answers for the tests with several correct answers	C, D, G	A, E	A, C, G	
Matching questions	1-F, 2-D, 3-A	1-E, 2-D, 3-A		

7. CELL CYCLE	1	2	3	4
Answers for the tests with one correct answer	C	A	B	B
Answers for the tests with several correct answers	A, D	A, D	E, G	
Matching questions	1-A, 2-D, 3-F	1-B,2-E, 3-C		

8. HEREDITY AND VARIABILITY	1	2	3	4
Answers for the tests with one correct answer	E	C	C	A
Answers for the tests with several correct answers	D,G	B, D, E	C, D, G	
Matching questions	1-C, 2-F, 3-D	1-D, 2-E, 3-A		

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ANSWERS

9. EVOLUTION	1	2	3	4
Answers for the tests with one correct answer	D	C	C	C
Answers for the tests with several correct answers	C, F, G	A, F, H	B, E, F	
Matching questions	1-E, 2-B, 3-F	1-D, 2-B, 3-A		

10. COORDINATION AND REGULATION	1	2	3	4
Answers for the tests with one correct answer	A	B	E	A
Answers for the tests with several correct answers	A, C, E	E, F	F, H	
Matching questions	1-B, 2-D, 3-C	1-C, 2-F, 3-E		

11. MOVEMENT	1	2	3	4
Answers for the tests with one correct answer	B	B	E	C
Answers for the tests with several correct answers	A, D, F	B, C, E	A, C, F	
Matching questions	1-B, 2-A, 3-D, 4-C	1-B, C, E, G 2-A, D, F, H		

12. BIOMEDICINE AND BIOINFORMATICS	1	2	3	4
Answers for the tests with one correct answer	C	B	D	D
Answers for the tests with several correct answers	A, D, G	-	-	
Matching questions	1-C, 2-E, 3-F	1-E, 2-F, 3-A		

13. BIOTECHNOLOGY	1	2	3	4
Answers for the tests with one correct answer	C	A	D	B
Answers for the tests with several correct answers	A, C, D	D, F, H	B, C, D	
Matching questions	1-E, 2-C, 3-B	1-E, 2-B, 3-C		

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GLOSSARY

A

Abdomen: The part of the body of a vertebrate containing the digestive and reproductive organs; the belly.

Absorption: uptake of substances by a tissue.

Acid: A chemical that reacts easily with other substances and turns litmus paper red; less than 7 on the pH scale

Acidification: the buildup of hydrogen cations, also called protons, reducing the pH;

Acne: inflammatory disease of the sebaceous gland

Actin: type of protein that forms microfilaments and responsible for many types of cell movements;

Action potential: short-term change of charge across cell membrane, in the result of moving of ions out and into of the cell;

Active immunity: is immunity in an organism resulting from its own production of antibody or lymphocytes

Active site: the part of an enzyme that interacts with the substrate during catalysis.

Active transport: transport of large molecules with energy loss.

Adaptation: a process that fits organisms to their environment.

Adenosine: a molecule consisting of an adenine and a ribose sugar;

Aerobic respiration: the process of breakdown organic molecules using oxygen.

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Aflatoxins: are poisonous carcinogens that are produced by certain molds (Aspergillus flavus and Aspergillus parasiticus) which grow in soil, decaying vegetation, hay, and grains;

Agglutination: adhesion of separate parts

Aging: the process of getting older;

AIDS: acquired immunodeficiency syndrome.

Albinism: Albinism is a range of disorders varying in severity. They are all caused by a reduction or absence of the pigment melanin, often causing white skin, light hair, and vision problems.

Allele: versions of gene for one trait.

Alveoli: tiny air sacs found in a lung which is used for gas exchange

Amino Acid: A molecule that joins with other amino acids to form proteins

Ammonia: a strongly basic compound of nitrogen and hydrogen that is very soluble in water and formed in nature by-product of protein metabolism.

Amylase: enzyme, which breaks down carbohydrates.

Anabolism: complex molecules are synthesised from simple ones;

Anaerobic respiration: respiration without oxygen;

Anaerobic respiration: the process of breakdown organic molecules without oxygen.

Analyze: Think about the different parts of a problem or situation to figure out how it is related to the whole.

Angiosperms: flowering plants

Antagonistic pair of muscles: a pair of muscles which have opposite functions.

Antibodies: are used by the immune system to destroy pathogens;

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Antidiuretic hormone: a hormone released by the posterior pituitary gland that increases the absorption of water by the kidney.

Antimicrobial resistance: the ability of a microorganism to stop an antimicrobial effect;

Aromatic rings: are hydrocarbons that contain benzene;

Arteriosclerosis: a disease of the arteries characterized by the deposition of fatty material on their inner walls.

Artery: Blood vessel that carries blood away from the heart

Arthropods: animals with segmented body, jointed limbs and exoskeleton

Asexual Reproduction: Reproduction that happens without sex; one organism creates one or more organisms

Atmosphere: a layer of gases which surrounds the Earth.

Atom: The smallest unit of a substance that has all of the properties of that substance

ATP: ‘energy currency’ in living things;

ATP synthase: an enzyme responsible for ATP production.

ATP yield: total number of ATP molecules synthesised during respiration;

Atrium: upper chamber of heart that receives blood from the body and lungs

Autosome: chromosome that carry genes for the body traits.

Auxin: plant hormone that regulates growth and other processes of plants.

B

B cells: a kind of lymphocyte that provide humoral immunity

Backbone: the row of connected bones that go down the middle of the back and

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protect the spinal cord.

Bacteria: The kingdom of life which has no cell membrane or nucleus and is always unicellular

Beriberi: a disease causing inflammation of the nerves and heart failure, ascribed to a deficiency of vitamin B1.

Bile: a yellow or greenish liquid that is made by the liver and that helps the body to digest fats.

Binomial nomenclature: a system of nomenclature in which each species is given a unique name that consists of a generic and a specific term.

Biofertilizer: a fertilizer which contains living microorganisms which help plant growth,

Biogeochemical cycle: the flow of chemical elements and compounds between living organisms and the environment.

Bioleaching: the extraction of metals from their ores by microbes;

Biological classification: Organization of how living things are related to each other

Biology: The study of living things

Biomechanics: a science that studies applying the basic principles of mechanics to living organism.

Bionics: the applying of biological methods of nature to the study and design of engineering technology.

Bioreactor: a device in which biological active substances are produced by iving organisms;

Biotechnology: using of biological processes or organisms to produce different products.

Blastula: early stage of development, two-layered ball like structure

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Body cell: All of the cells in an organism not involved in reproduction

Bond: An electrical force that links atoms together

Brain: organ that serves as the center of the nervous system in all vertebrate and most invertebrate animals.

Brain-Computer Interfaces: a system for exchanging information between the brain and an electronic device.

Bronchi: major air passages of the lungs which diverge from the windpipe.

Bundle of His: the bundle of cardiac muscle fibers that conducts the electrical impulses which regulate the heartbeat.

C

Calorie: Measure of energy, usually contained in food

Cancer: Disease where cells grow in an uncontrolled way

Canines: a pointed tooth between the incisors and premolars

Canines: a pointed tooth between the incisors and premolars of a mammal.

Capillaries are blood vessels that connect arteries and veins

Carbohydrate: An essential chemical in all cells that is broken down to form sugars; glucose, sucrose, lactose, galactose

Carbon Dioxide: Molecule made up of one carbon and two oxygens, produced by animals and other organisms; main contributor to manmade global warming

Carbon: a widely distributed element that forms organic compounds in combination with hydrogen, oxygen, etc.

Carcinogen: any chemical or physical factor, that cause cancer.

Carrying capacity: the number of organisms which a region can support without environmental degradation.

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Catabolism: complex molecules are degraded to simple ones;

Catalyst: a substance that causes or accelerates a chemical reaction.

Cell: the smallest unit of life;

Cell cycle: a life of a cell from one cell division to the next one;

Cell cycle: it a period of cell life from one cell division to the next cell division.

Cell differentiation: A process where cells change to perform different roles

Cell division: A process where one cell becomes more than one cell

Cell membrane: Surrounds the cell and keeps it together; also decides what material enters and leaves

Cell organelle: Parts of the cell that perform specific functions

Cell wall: The stiff outer layer of a cell that protects the cell and gives it shape

Cell: The basic structural and functional unit of all organisms

Cellular respiration: the process of breakdown organic molecules producing energy.

Cellular respiration: The process where organisms get energy from organic molecules

Cellulose: a substance that is the main part of the cell walls of plants and that is used in making various products (such as paper)

Centrosome: an organelle, which plays role in cell division.

Cephalosporins: are a class of β-lactam antibiotics originally derived from the fungus Acremonium, which was previously known as “Cephalosporium”;

Characteristic: A distinguishing quality of something; generosity is one of his best characteristics

Checkpoint: a particular moment in the life of a cell, at which cells “decide” to

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continue cell cycle or do not;

Chemical potential energy: absorbed or released energy during chemical reaction;

Chemiosmosis: is the movement of ions across a membrane, down their concentration gradient;

Chest cavity: internal space of chest where heart and lungs are located.

Chitin: a kind of polysaccharide which is found in the structure of fungi cell walls and insect skeleton.

Chlorinated hydrocarbons: are very potent nerve toxins to insects, and their initial use led to significant improvements in insect control;

Chloroplast: Organelle in plants and some other organisms which is responsible for photosynthesis

Cholinergic synapses: synapses that use the acetylcholine as a neurotransmitter.

Chromatin: a fluid inside a nucleus, consisting of DNA, proteins and other molecules.

Cilia: a short structure, used for movement.

Cladogram: a branching diagram showing the cladistic relationship between a number of species;

Classification: systematic arrangement in groups or categories according to established criteria.

Cloning genes: process of isolation of needed fragment of DNA and making many copies of the fragment in vitro;

Closed circulatory system: blood is contained inside blood vessels, circulating unidirectionally.

Codominance: when the neither phenotype is dominant, the individual expresses both phenotypes.

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Cohesion: the sticking together of particles of the same substance;

Compound: A chemical combination of two or more atoms (of different elements)

Concentration gradient: gradual change in the concentration of solutes present in a solution between two regions;

Consumer: Living thing that eats other living things

Cotyledon: part of the seed that will grow into the leafs.

Covalent Bond: A type of chemical bond where electrons are shared between the atoms

Cristae: folds of inner membrane of mitochondria:

Crop: organ used for storage of food in birds.

Crossing over: exchange of pieces between non-sister chromatids.

Cystitis: inflammation of urinary bladder or urethra by infection of bacteria.

Cytokinesis: the division of the cytoplasm.

D

Decomposer: A type of living thing that survives by consuming dead organic matter

Deforestation: the cutting down of trees without replacing them.

Denaturation: process of altering the shape of a protein without breaking the peptide bonds that from primary structure.

Dentine: hard dense bony tissue forming the bulk of a tooth, under the enamel.

Dermis: thick layer of skin under epidermis

Development: growth from one cell to multicellular organism

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Dialysate: the liquid into which material passes by way of the membrane in dialysis

Diaphragm: a large flat muscle that separates the lungs from the stomach area and that is used in breathing.

Diarrhea: an intestinal disorder that causes pass waste from your body very frequently and in liquid state

Dicot: plant with two cotyledons(seed leaves)

Differentiation: Process where cells grow and become different than their mother cell

Diffusion: movement of substances from the region of higher concentration to the area of lower concentration;

Digestion: breaking down big molecules into small molecules.

Disaccharide: a double sugar composed of two molecules of monosaccharides linked.

DNA: double-stranded nucleic acid that contains the genetic information;

DNA replication: process by which DNA makes a copy of itself when cell divides.

DNA: a molecule that store and transmit genetic information.

DNA: Abbreviation for deoxyribonucleic acid, which contains the genetic instructions for all forms of life

Dominant: A genetic characteristic that is always expressed by the organism

Donor: a person who gives some of their blood or a part of their body to help someone.

Double helix: structure formed by two strands of DNA.

Dynamic work of muscles: muscle contract and relax to move different parts of your body.

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E

Ectoderm: the outermost layer of cells or tissue of an embryo in early development

Eczema: inflamed skin characterized by redness, itching, and scaly.

Edema: a condition characterized by an excess of watery fluid collecting in the cavities or tissues of the body.

Egg cell: female reproductive cell.

Electrocardiography: the process of recording the electrical activity of the heart over a period of time using electrodes. (ECG)

Electrocytes: a modified muscle or nerve cell that generates electricity.

Electron carriers: accept electrons from one molecule and donate them to another molecule;

Electroreception: the biological ability to perceive natural electrical stimuli.

Electroreceptors: cells capable of detecting electric fields.

Elimination: the passing of undigested material out of the body.

Embryo: an unborn baby less than eight weeks.

Embryology: The study of how organisms develop; the more closely related two organisms are, the more similar they are as they develop

Emulsification: the breakdown of large fat droplets into smaller droplets.

Enamel: the white, compact, and very hard substance covering and protecting the dentin of a tooth.

Endoderm: the innermost layer of cells or tissue of an embryo in early development.

Endoplasmic reticulum: a network of membranous tubules in the cytoplasm of

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cell.

Endoplasmic reticulum: Organelle that produces proteins and fats

Energy: the ability to do work.

Energy: The property of something's ability to do work

Enuresis: frequent inability to control urination known as bedwetting.

Environment: The complex of physical, chemical, and biotic factors (e.g., climate, soil, living things) that act upon an organism or an ecological community and ultimately determine their forms and survival

Eon: a major division of geological time, subdivided into eras;

Epidermis: outermost layer of the skin

Era: a major division of time that is a subdivision of an eon.

Erosion: the wearing away of land by different forces (water, wind, ice, living organism, temperature) cause.

Erythrocyte: are red blood cells, which carry oxygen

Esophagus: the part of the digestive system which connects the throat to the stomach.

Ethical question: involve consideration of conflicting moral choices and dilemmas;

Eukaryote: A type of organism that has a true nucleus in its cell(s)

Evolution: the change in groups of organisms over a long period of time.

Excretion: process of elimination of wastes from the body

Excretory system: a system that removes excess water and unnecessary products. It consists of kidneys, ureter, urinary bladder, and urethra.

Exhalation: movement of air out of lungs.

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Exoskeleton: external skeleton that supports and protects an animal’s body.

Exponential growth: development at an increasingly rapid rate in proportion to the growing total number or size.

F

Facilitated diffusion: molecules diffuse through the cell membrane with the help of special proteins;

Fast-twitch fibers: muscle fiber that contracts relatively rapidly, utilized especially in actions requiring the maximum effort of short duration;

Fern: seedless, nonflowering vascular plant which lives in tropical to temperate regions.

Fetus: the unborn embryo from its eighth week of development till its birth.

Fiber: a thin, threadlike structure.

Fibrous proteins: composed of long linear polypeptide chains that are bundled together to form strand or sheets.

Filament: group of myofibrils;

Filtrate: a solution produced from filtration .

Filtration: renal process whereby fluid in the blood is filtered across the capillaries of the glomerulus and into the urinary space of Bowman’s capsule;

Fixed joint- joint between two bones that doesn’t move

Flagellum: a long structure, used for movement.

Fluid: Anything that flows; both gases and liquids are considered fluids

Food poisoning: becoming ill after eating spoiled food

Forensics: science that use of scientific knowledge or methods in solving crimes

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Fossils: the remains of organisms which lived long time ago.

Fraternal twins: twins those are developed from two zygotes.

Fungi: The kingdom of living things that are eukaryotic and make their own energy; mushrooms, yeast, molds

G

Galvanometer: an instrument for detecting and measuring small electric currents.

Gametes: The cells that are responsible for sexual reproduction; sperm, eggs

Gametogenesis: process by which gametes are produced.

Gene: a segment of DNA which codes a particular protein.

Genetic drift: is the change in the frequency of an existing allele in a population due to random sampling of organisms.

Genetic engineering: constructing new DNA from DNA of different species.

Genetic marker: is a gene or DNA sequence with a known location on a chromosome that can be used to identify individuals or species;

Genetic variation is the difference or diversity in gene alleles in a population;

Germ: A small organism that causes disease

Gills: respiratory organ of aquatic animals.

Gizzard: a muscular, thick-walled part of a bird’s stomach for grinding food.

Glia: cell that feeds and protects the nerve cell.

Globular proteins: coiled into compact shapes with hydrophilic outer surface make them water soluble;

Glycolysis: splitting of glucose molecule into two pyruvate molecules

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Glycolytic fibers: muscle fibers primarily use glycolysis to generate ATP;

Glycosidic linkage: is a type of covalent bond, formed between monosaccharides.

GMO: genetically modified organisms.

Goggles: A safety device used whenever the eyes could be injured by a chemical or physical experiment

Golgi apparatus: Organelle that packages proteins and fats so that they can leave the cell

Greenhouse effect: the Sun’s radiation is trapped in the atmosphere and leads to a warming of the Earth.

Greenhouse gas: a gas in the atmosphere which can trap the heat escaping from Earth.

Growth factors: compounds that are capable of stimulation of growth, division and differentiation of cells;

Growth medium: is a medium on which microbes grow.

Growth: The increase of size of an individual; can also be the presence of something

Gymnosperm: a group of plant that have seeds unprotected by an ovary or fruit, including the conifers, cycads, and ginkgo.

H

Heart attack: blocking of heart arteries which brings to death of heart muscles

Heart automaticity: the ability of the heart cells to self-excitation without any external influences.

Heart conduction system: controls the heart rate and generates electrical impulses which conducts them throughout the muscle of the heart, stimulating the heart to contract.

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Hemoglobin: reddish-brown pigment, which give blood its color and carry oxygen;

Heredity: the passing of a trait from parents to offspring.

Heterotroph: Living thing that needs to consume other organisms for food

heterozygous appear intermediate trait.

HIV: human immunodeficiency virus.

Homeostasis: Maintenance of a constant internal environment in an organism.

Homologous chromosomes (also called homologs): a pair of chromosomes in diploid cells. They have same shapes, sizes, and genes. Each of them comes from a different species;

Homology: similarity resulting from common ancestor.

Humoral immunity: antibody-mediated immunity

Humoral regulation: regulation by the endocrine system.

Hybrid is an offspring resulting from cross-breeding;

Hydrolysis: the cleavage of chemical bonds by the addition of water.

Hydrophilic: water loving, having a strong affinity for water;

Hydrophobic: water fearing, tending not to dissolve in, mix with, or be wetted by water.

Hypertension: high blood pressure

Hypodermis: the deepest part of the skin which contains fat cells

I

Identical twins: twins that are produced from a single zygote.

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Immunity: ability to completely fight off disease

Incisor: tooth at the front of the mouth, adapted for cutting and biting.

Incomplete dominance: when both alleles for the trait have same level of affect.

Inflammation: physical condition in which part of the body becomes reddened, swollen, hot, and often painful.

Ingestion: the process of taking food, drink, or another substance into the body.

Inhibitor: a substance which slows down or prevents a particular chemical reaction or other process.

Intercourse: physical sexual contact between individuals.

Interphase: a part of a cell cycle, when cell is not dividing.

J

Joint: the area where two bones are attached for the purpose of permitting body parts to move.

K

Kidney stones: formation of stones that block urine flow caused mostly by unhealthy diet.

Kidney transplant: a surgical procedure to remove a healthy, functioning kidney from a living or brain-dead donor and implant it into a patient with non-functioning kidneys.

Kidney: either of a pair of bean-shaped organs in the back part of the abdominal cavity that forms and excrete urine, regulate fluid and act as endocrine glands.

Kingdom: Most general classification of living things (sometimes placed under “Domains”)

Krebs cycle: is a cycle of biochemical reactions, which produces CO2, NADH,

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FADH2 and ATP.

L

Lactate: the end by-product of lactic acid fermentation;

Large intestine: end part of the intestine that is wider and shorter than the small intestine.

Leukemia: Cancer of the blood or bone marrow

Leukocyte: white blood cells, which protect the organism

Lichen: A combination of a fungus and an algae that help each other

Life cycle: The series of stages in form and functional activity through which an organism passes between origin and expiration

Limbs: an arm or leg of a person or four-legged animal, or a bird’s wing.

Linear magnification: the ratio of image length to object length measured in planes.

Lipase: enzyme, which breaks down lipids.

Lipids: biomolecules that are soluble in organic solvents and insoluble in water

Liquid: The state of matter where the particles are loose and form the shape of their container but do not necessarily fill up the container

Living thing: An organism

Lymph: fluid that circulates through lymphatic system

Lymphatic system: network of tissues and organs that help rid the body of toxins, waste and other unwanted materials.

Lymphocyte: A lymphocyte is a type of white blood cell that is part of the immune system. There are two main

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Lysosome: an organelle in the cytoplasm of cells containing degradative enzymes.

Lysosome: Organelle that digests food and waste in the cell

M

Malignant tumor: a group of cancer cells, which can travel throughout the body via blood or lymph vessels;

Matrix: fluid which fills the mitochondria;

Mechanoreceptors: sensory receptor that responds to mechanical pressure

Membrane potential: the difference between charge outside and inside of the cell.

messenger RNA: a type of RNA that convey genetic information from DNA to the ribosome, where they specify the amino acid sequence of the protein;

Micrograph: a photograph of an image under the microscope.

Mitochondrion: is an oval-shaped organelle, that is ‘power house of the cell’;

Mitosis: a type of cell division;

Monosaccharide: a simple sugar, the basic unit of carbohydrates;

Multicellular: many celled organism.

Muscle contractions: a process in which muscles become short, moving our body.

Muscle fatigue: decrease in muscle ability to generate force, because of tiredness.

Mutagenesis is a process by which the genetic information of an organism is changed, resulting in a mutation;

Mutation is a change in DNA of an organism;

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Myelin sheath: a lipid-rich substance that surrounds the axon of some nerve cells

Myofibril: a contractile fibril of skeletal muscle, composed mainly of actin and myosin;

Myoglobin: store oxygen in tissues;

Myosin: the protein that converts chemical energy to the mechanical energy and cause the muscle contraction.

N

Natural selection: the process whereby organisms better adapted to their environment tend to survive and produce more offspring.

Nausea: sickness at the stomach, an involuntary impulse to vomit an atom; fission, fusion

Nephron: a unit of filtration in a kidney that regulates water balance and removes toxins.

Neural regulation: regulation by nervous system.

Neuron: a nerve cell.

Neuroprosthetic: Any biomedical engineered device designed to be linked to the peripheral or central nervous system and enhance the cognitive, motor, or sensory abilities of an organism.

Neurotransmitters: chemicals that carries messages between neurons or between neurons and muscles.

Ninhydrin reagent: chemical that reacts with free amino groups of amino acids.

Nitrogen fixation: is a process by which nitrogen in the Earth’s atmosphere is converted into ammonia (NH_3) or other molecules available to living organisms.

Nitrogen: a chemical element that is a gas with no colour or taste, forms most of the earth’s atmosphere and is a part of all living things.

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Nonreducing sugar: sugar that cannot donate electrons to other molecules and therefore cannot act as a reducing agent;

Nonvascular plant: plants without vascular tissue such as mosses.

Nucleic acid: In the nucleus of a cell, there are two major types of nucleic acids: DNA and RNA

Nucleotide: the basic building block of nucleic acid polymers; an organic compound made up of a nitrogenous base, a sugar, and a phosphate group.

Nucleotide: monomer of DNA that consists of sugar, phosphate group, and nitrogenous base.

Nucleotides: These are found on a strand of DNA or RNA as a sequence of bases

Nucleus: In biology, this refers to the middle of a cell; in physical science, this refers to the center of an atom

Nutrient: Any molecule that is needed for an organism to survive

O

Objective: Based upon fact

Observation: Noticing or paying attention

Offspring: Children, of any organism

Open circulatory system: organisms with open circulatory system have open ended vessels. Blood leaves the vessels and enters to the body space.

Order: Classification of living things between Class and Family

Organ system: Group of organs that together perform a common function

Organ: Group of tissues that perform a certain function

Organelle: Part of a cell that performs a function for the cell

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Organic certification: is a certification process for producers of organic food and other organic agricultural products;

Organic molecule: A molecule that contains carbon atoms bonded together

Organic: Comes from living things

Organism: A living thing that can live and reproduce independently

Organogenesis: formation of organs

Origin of replication: starting point of DNA replication;

Osmoreceptor: any of a group of cells sensitive to rise in the solute concentration of the blood;

Ovary: organ that produces female gametes and female sex hormones.

Oxidation-reduction reactions: involves transfer of electrons between two chemical substances;

Oxidative fibers: muscle fibers primarily use aerobic respiration to generate ATP;

Oxidative phosphorylation: is the reaction in which cells use enzymes to oxidize nutrients, by releasing energy which is used to produce ATP.

Oxidizing agent: oxidises other substances and gains electrons.

Oxygen saturation: the ratio of the amount of oxygen bound to the hemoglobin

P

Pacinian corpuscles: one type of mechanoreceptor cell which are sensitive to vibration and pressure.

Partial pressure: a measuring of oxygen concentration;

Passive immunity: the immunity resulting from the injection of antibodies or lymphocytes from another organism

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Passive transport: transport of small molecules without energy loss.

PCR: polymerase chain reaction which is used to make many copies of DNA molecule fragment;

Pepsin: enzyme, which breaks down proteins.

Peristalsis: the involuntary constriction and relaxation of the muscles of the intestine or another canal, creating wave-like movements which push the contents of the canal forward.

Permanent teeth: second set of teeth in mammals that grows as the milk teeth are shed

Phagocytes: leukocytes which digest viruses and bacteria

Phosphate bonds: high energy containing bonds.

Phosphorylation: addition of phosphoryl group during the reaction

Photosynthesis: process by which plant convert energy from sunlight into chemical energy.

Photosynthesis: The process that happens in plants and some other organisms which takes the sun's energy and turns it into usable energy; $6CO2 + 6H2O + Light = C6H12O6 (glucose) + 6O2$

Phylogenetic tree: a diagram showing the evolutionary interrelations of a group of organisms derived from a common ancestral form;

Placenta: connects the developing fetus to the uterine wall to allow nutrient uptake, via the mother's blood supply during pregnancy.

Plant hormones: organic substances that control plant development and growth.

Plasma: liquid part of blood.

Plasmid DNA: is a small DNA molecule within a cell that is physically separated from a chromosomal DNA;

Plasmid: ring-shaped DNA molecule of bacteria.

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pollen: the fertilizing element of flowering plants, consisting of fine, powdery, yellowish grains or spores, sometimes in masses.

Pollination: the transfer of pollen from the anther to the stigma.

Pollution: contamination of environment by rubbish, oil spill and so on.

Polymer: large molecule or macromolecule composed of many repeated parts

Polyploidy: organism which contain more than two sets of chromosomes.

Polysaccharides: a large molecule composed of a long chains of monosaccharide units bound together by glycosidic linkage.

Population: a group of organisms of one species that live together.

Positive geotropism: growth of plants towards the gravity.

Premolar: situated in front of the molar teeth.

Producer: Living thing that makes its own energy from the sun

Prokaryote: A type of living thing that is single-celled and has no true nucleus

Protein: A sequence of amino acids

Protist: Kingdom of life made up of single-celled eukaryotes

Pulmonary circulation: movement of blood from the heart to the lungs for oxygenation, then back to the heart again.

Pulp cavity: the space within a tooth that contains the pulp.

Pulp: soft part of the tooth where blood vessels and nerves are found.

Pulse: the regular beating of the heart, especially when it is felt at the wrist or side of the neck.

Purkinje fibers: networks of fibers that receive conductive signals originating at the atrioventricular node, and simultaneously activate the left and right ventricles.

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Pyruvate oxidation: conversion of pyruvate into acetyl-CoA

R

Reabsorption: renal process that returns filtered water and nearly all major nutrients to blood;

Reaction: When one or more substances are changed into other substances

Recombinant DNA: a molecule of DNA which is produced from DNA of different organisms;

Recombination is shuffling existing alleles to produce new combinations of genes.

Reducing agent: reduces other substances and loses electrons;

Reducing sugar: sugar that serves as a reducing agent due to its aldehyde or ketone group in its molecular structure;

Refractory period: a period immediately during which a nerve or muscle is unresponsive to further stimulation.

Regeneration: The process that creates something over again

Replication: is a process of DNA copying.

Reproduce: To create more of

Reproduction: The process of creating offspring

Research: Discovering information that other scientists have already published

Resistance: (Biology) Ability to fight off some amount of disease

Resource: A supply of something that can be used when needed

Respiratory minute volume: the amount of inhaled or exhaled air in a minute.

Respiratory rate: rate of breathing

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Resting potential: the resting state of a cell, when outside is positive, inside is negative.

Restriction enzymes: enzymes that cut DNA at specific sites.

Restriction site: are locations on a DNA molecule containing specific sequences of nucleotides, which are recognized by restriction enzymes;

Ribosomal RNA: component of ribosome functioning in protein synthesis.

Ribosome: sphere-shaped organelle composed of protein and rRNA that serves as the site of protein synthesis;

RNA: single-stranded nucleic acid that function in protein synthesis.

Robotics: an applied science that includes mechanical engineering, electrical engineering, computer science and developing automated technical systems.

S

Salivary glands: glands which produce saliva

Saltatory conduction: is the propagation of action potentials along myelinated axon from one node of Ranvier to the next node, increasing the conduction velocity of action potentials

Sarcomere: structural unit of the myofibril where structural and contractile proteins are organized in a specific sequence.

Saturated: Completely full; for a solution, the most amount of solute that can be put into the solvent

Scabies: skin disease with itching and small raised red spots, caused by the itch mite.

Sebaceous gland: (oil gland)a small gland in the skin which secretes a lubricating oily matter (sebum) into the hair follicles to lubricate the skin and hair.

Secondary sex characteristics: sex characteristics that appeared during puberty.

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Secretion: renal process that removes additional wastes from the blood and adds them to the filtrate.

Selective breeding: intentional breeding of animals or crops with particular characteristics;

Selectively permeable: crossing of the cell membrane by some molecules.

Semi-movable joints: joints between bones in which the motion is limited

Senescence cells: our cells, that can not divide or die; when human becomes older their number increases.

Sex chromosomes: chromosome that carry genes which have a role in sex determination.

Sexual reproduction: The combination of two individuals (genetically) to form one or more new organisms

Sexually transmitted disease: an infection that can be transferred from one person to another through sexual contact.

Sigmoid curve: a mathematical function having a characteristic “S”-shaped curve.

Sink: plant organ that store sugar.

Slow-twitch fibers: muscle fiber that contracts relatively slowly and is resistant to fatigue.

Small intestine: narrow upper part of the intestine where digestion of food completes and absorption starts

Solute: substance dissolved in a solution.

Solution: homogeneous mixture of two or more substances.

Solvent: able to dissolve other substances.

Speciation: the formation of new and distinct species.

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Species: the group of organisms with similar features, which can interbreed and produce viable and fertile offspring;

Specific heat capacity: the amount of heat per unit mass required to raise the temperature by one degree Celsius;

Sperm cell: male reproductive cell.

Spinal cord: cord of nerve tissue located within the vertebral column.

Spindle fibers: microtubules, which play important role in mitosis;

Spleen: an abdominal organ involved in the production and removal of blood cells in most vertebrates and forming part of the immune system.

Starch: A kind of storage polysaccharide found in plants.

Static work of muscles: muscles stay contracted to hold your body in certain position.

Stem cell: A type of cell that can turn into any other type of cell.

Sternum: Flat bone that connects ribs to each other forming the front part of rib cage.

Sticky ends: short lengths of unpaired bases;

Stimulus: Anything that affects an organism

Stomach: saclike organ which store and digest food

Stomata: pores on the underside of the leaf which accelerate gas exchange and transpiration.

Stroke: death of brain tissue occurred as a result of artery blockage

Struggle for existence: competition between organisms of a population;

Substance: A type of matter that has the same properties; water, oxygen, carbon dioxide, diamond

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Substrate: the substance on which an enzyme acts.

Surface area: the amount of surface or the outer surface that the cells or organisms have.

Synapse: a junction between two nerve cells, consisting of a gap across which impulses pass by diffusion of a neurotransmitter.

Systemic circulation: movement of blood from the heart through the body to provide oxygen and nutrients to the tissues of the body while bringing deoxygenated blood back to the heart.

T

T cells: leukocytes that provide cell-mediated immunity

T tubules: small tubule which run transversely through a striated muscle fiber and through which electrical impulses are transmitted from the sarcoplasm to the fiber’s interior.

Tagged primer: tag is a molecule that is attached chemically to primer in order to aid in the labeling and detection of a biomolecule.

Taxon: is a group of one or more populations of an organism or organisms seen by taxonomists to form a unit.

Telomere: a fragment of DNA at the end of each chromosome, that protects genes from being damaged after the replication;

Testcross: cross which is made for to determine dominant trait genotype in organism by using recessive gene.

Testes: organs that produce male gametes and male sex hormones.

Tetter: any of various eruptive skin diseases

Theory: one explanation for different facts and observations.

Thermoregulation: the control of body temperature.

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Threshold: a particular value which action potential have to reach.

Tidal volume: volume of gas inhaled or exhaled in each respiration, during a normal, regular breathing

Tissue fluid: fluid between cells

Tissue: Group of cells that perform a similar function

Tonoplast: the membrane that is surrounds plant vacuole.

Tonsil: either of two small masses of lymphoid tissue in the throat, one on each side of the root of the tongue.

Toxin: a poisonous substance and especially one that is produced by a living thing

Trachea: a large membranous tube reinforced by rings of cartilage, extending from the larynx to the bronchial tubes and conveying air to and from the lungs; the windpipe.

Trait: characteristic of an organism.

Transfer RNA: a cloverleaf shaped RNA that transports specific amino acid to the ribosome;

Transgenic organism: an organism which contains DNA form another organism, and also called genetically modified organism.

Translocation: movement of organic compounds inside of the plant.

Transpiration: the loss of water as vapor.

Triglyceride: an energy-rich compound made up of a single molecule of glycerol and three molecules of fatty acid.

Tropomyosin: a rod-shaped protein attached to actin in a regularly repeating sequence;

Troponin: a component protein that is associated with each actin-tropomyosin complex;

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Typhoid: an infectious bacterial fever with an eruption of red spots on the chest and abdomen and severe intestinal irritation.

U

Ultraviolet: A form of electromagnetic radiation that has more energy than visible light; most ultraviolet light is usually blocked in our atmosphere by ozone

Unicellular: Made up of one cell

Urea: a substance found in urine and also made from ammonia,

Urea: a water-soluble organic compound, synthesized from ammonia and carbon, formed by the metabolism of proteins and excreted in the urine.

Ureter: a muscular duct or tube conveying the urine from a kidney to the bladder or cloaca

Urethra: the duct by which urine is conveyed out of the body from the bladder, and which in male vertebrates also conveys semen.

Uric acid: an almost insoluble compound a breakdown product of nitrogenous metabolism.

Urinary bladder: a membranous sac in many vertebrates that serves for the temporary retention of urine and discharges by the urethra.

Urine: a mixture removed from our body, it consists of water, urea, salts and other excretory substances.

Urine: liquid waste excreted by the kidneys, in humans being a yellowish, slightly acid

V

Vaccine: immunobiological drug, which is used to produce immunity against infectious diseases.

Vacuole: Organelle that stores nutrients in the cell

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Variation: differences among organisms in one population.

Variation: variety of traits in species.

Vascular: relating to or denoting the plant tissues (xylem and phloem) which conduct water, sap, and nutrients in flowering plants, ferns, and their relatives.

Vector: delivers recombinant DNA to the cells of the organism.

Veins: are blood vessels that transport blood to the heart

Vena cava: any of the large veins by which in air-breathing vertebrates the blood is returned to the right atrium of the heart.

Ventricle: one of two sections of the heart that pump blood out to the body.

Vesicle: saclike membranous structure that store and transport cellular products.

Vessel: tube shaped structure that carries blood in the body.

Virus: A small particle that contains DNA or RNA and is able to reproduce only inside of a living cell

Vital capacity: greatest amount of air that can be forced from the lungs after maximum inhalation

Vitamin: organic molecule essential for body processes

Volume: a space taken up by a cell or an organism.

W

Wax: an oily organic compound insoluble in water but soluble in organic solvents; essential structural component of living cells (along with proteins and carbohydrates).

Y

Yeast: an organism of the kingdom Fungi lacking chlorophyll and feeding on

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organic matter; ranging from unicellular or multicellular organisms to spore-bearing syncytia.

Z

Zygote: fertilized egg cell which is formed from the union of female egg cell with a male sperm cell.

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BIOLOGY

Grade 10

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